

Complete Certificate Closure for a Constraint-Native Gravity–Quantum Realization under AASC

Modules A–D Closure Edition: Metric Sources, Common Mass-Charge Matching, Upstream Continuation, Exact Certificate-Fibre Exhaustion, and Unique Role-Occupancy under Frozen FEP-1.0

Amos Jay Maley
2026

Frozen dependency lock (FEP-1.0).

$$\text{TargetAdeq}_D \implies K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D) \implies A_D^+ \implies I_D.$$

Gravity and quantum standing enter only after the singleton realized physical interior; Einstein and Schrödinger dynamics are downstream bookkeeping projections. Modules A–D are later construction, matching, continuation, and certificate-completion theorem objects. No source tensor, charge, mass readout, clock, history, tensor colimit, skin quotient, live residue, assignment orbit, or realization conclusion may authorize an upstream role.

Accepted modular imports. **Module A** supplies attachment-safe metric-source representation, Ward–divergence correspondence, and finite composite closure. **Module B** supplies the attached total-EMT charge, exact zero-transfer active readout, Wigner-to-Bargmann contraction, one projection-wide inertial–active factor, and calibrated equality in its frozen stable scalar envelope. **Module C** supplies an independently formed fixed-clock RPI continuation category, metric/quantum/record/compatibility descent, directed exchange, and a coherent sixteen-vertex continuation cube. **Module D** supplies source-locked tensor completion, the induced joint probe and maximal lawful skin kernel, exact eight-fibre exhaustion, live-residue closure, one lawful assignment orbit, and certificate-relative realization licensure. All artifact identities and positive independent reviews are embedded in the release package.

Frozen-gate status after Modules A–D. G1–G5 and G7 are closed in the exact accepted module envelopes. G6 remains a separately correct conditional implication under TB1–TB11. The core FEP-1.0 complete-certificate realization endpoint is therefore closed relative to the frozen target; the manuscript does not claim uniqueness across different source, state, seed, potential, or matching targets, strong equivalence, or all quantum-gravity dynamics.

Abstract

This manuscript is the Modules A–D closure edition of the frozen capstone program for a constraint-native gravity–quantum realization under the Admissibility and Standing Constraint framework. The dependency order remains

$$\text{TargetAdeq}_D \Rightarrow K_D \Rightarrow A_D^+ \Rightarrow I_D,$$

with gravity and quantum standing downstream of the singleton realized physical interior, and Einstein and Schrödinger dynamics downstream bookkeeping projections. No metric tensor, source distribution, Poincaré charge, Hamiltonian mass coefficient, clock, continuation seed, tensor colimit, skin relation, occupation orbit, or realization certificate is permitted to authorize an upstream role.

Accepted Module A [1, 2] constructs the genuine source-bearing subtype, the structured source-witness quotient, and the attachment-safe natural representation

$$q_g : \widehat{Q}_{\text{src}} \longrightarrow \text{Src}_g^{\text{dist}} / \simeq_{\text{law}}.$$

It proves unique tensor-density representation, lawful quotient descent, the Ward–divergence equivalence, and finite completed-composite closure with exact term coverage and signed internal-exchange cancellation.

Accepted Module B [3, 4] operates on the operator-presentable stable scalar matching subtype. A normalized-ray family of concrete Module A sources is polarized to reconstruct one attached operator total EMT. Its translation charge gives the stable Poincaré Casimir, its total-EMT form factor obeys $A(0) = 1$, and an explicit Wigner-to-Bargmann contraction yields

$$H_{\text{kin}} = -\frac{\hbar^2}{2m_{\text{Cas}}} \Delta.$$

The scalar readouts have distinct downstream types but share one attached charge line. In the frozen envelope,

$$m_g(x) = \alpha_D m_i(x), \quad \alpha_D = \frac{\kappa c^2}{4\pi\beta_g} > 0,$$

and common calibration gives $m_g = m_i = m_{\text{Cas}}$.

Accepted Module C [5, 6] constructs the upstream continuation before any downstream solution is used. A certified local clock and smooth RPI-level seed form a fixed-clock Moore-history category. Lawful metric/source, full-ray quantum, record, and compatibility processes descend from that one upstream history; the metric image intertwines with the fixed Einstein–Module-A Cauchy development and the quantum image is Wigner–Stone–Schrödinger classified. Closed prediction and actual exchange-extended continuation are distinct maps. One common incidence object restricts to directed role exchanges satisfying the source-target cocycle, and the four directions generate a genuine sixteen-vertex continuation cube. The abstract full-ray carrier uses its standard smooth projective-Hilbert-manifold structure only as regular local chart structure, and the cube coherence is whiskered coordinatewise pasting.

Accepted Module D [7, 8] closes the remaining certificate and occupancy burden. It separates source-locked primary tensor interfaces from their lawful covariant completions, constructs those completions by explicit left Kan path terms in the covariant slice

$$\text{TenInt}_D = \text{Set}^{S_D} / \mathcal{W}_D,$$

and forms the total tensor ledger as the colimit of the completed generator/overlap diagram. The accepted probes extend by the same adjunction and induce the tensor probe; the full joint probe

is then formed, and its lawful kernel pair is the unique maximal standing-neutral skin congruence. Only afterward is the complete eight-vertex target fixed. Module D proves candidate, quotient, role-set, eligibility, compatibility, live-residue, and failure completeness; eight target-relative fibre-finality theorems; one compatible bijective assignment; and one lawful orbit:

$$|\Sigma_D^{gg}/\Gamma_D^{\text{lawful}}| = 1.$$

The imported Role-Occupancy Closure theorem therefore licenses one complete constraint-native realization class of the exact frozen target, unique up to lawful equivalence.

The integrated endpoint is certificate-relative. Different source systems, potentials, initial data, initial rays, RPI seeds, or matching packages define different target instances and require new certificate analyses. The conditional TB1–TB11 weak-equivalence implication remains separate: under its explicit small-body hypotheses, equal initial metric data determine the same local geodesic. The manuscript does not claim strong equivalence, generic curved-spacetime mass, a nonzero interacting quantum-gravity law, or all quantum-gravity dynamics.

Read This First: Modular Proof Spine

$$\text{TargetAdeq}_D \Longrightarrow K_D \Longrightarrow A_D^+ \Longrightarrow I_D,$$

then

$$I_D \Longrightarrow \begin{cases} \text{AERGravity}_D \Longrightarrow \Pi_g \Longrightarrow \text{Einstein}_D, \\ \text{QuantumStand}_D \Longrightarrow \Pi_H \Longrightarrow \text{Schrodinger}_D, \end{cases}$$

then accepted Module A

$$Q_{\text{resp}} \Longrightarrow Q_{\text{resp}}^{\text{src}} \Longrightarrow \hat{Q}_{\text{src}} \xrightarrow{q_g} \text{Src}_g^{\text{dist}} / \simeq_{\text{law}} \Longrightarrow \nabla_\mu \mathbf{T}^\mu{}_\nu = J_\nu,$$

accepted Module B

$$\hat{Q}_{\text{mass}}^{\text{op}} \Longrightarrow \hat{\mathbf{T}}_x \Longrightarrow m_{\text{Cas}} \Longrightarrow \begin{cases} m_g = \alpha_D m_{\text{Cas}}, \\ m_{\text{i}} = m_{\text{Cas}}, \end{cases}$$

accepted Module C

$$\text{RPI seed} \Longrightarrow \mathbf{Cont}_D^\tau,$$

$$\mathbf{Cont}_D^\tau \Longrightarrow \text{four faithful role processes},$$

$$\Longrightarrow \text{strict or exchange-certified four-cube}.$$

and accepted Module D

$$\text{source-locked primary interfaces} \Longrightarrow \text{lawful cosieve completions} \Longrightarrow L_D^0,$$

$$L_D^0 \Longrightarrow S_D^0 \Longrightarrow \text{Residue}_D^{\text{live}},$$

$$\text{Residue}_D^{\text{live}} \Longrightarrow |\Sigma_D^{gg}/\Gamma_D^{\text{lawful}}| = 1 \Longrightarrow \text{RealizationLicensed}(C_D^{ABC}, T_D^{gg}).$$

Current proof-gate ledger

Gate	Status in v1.6.0	Exact basis
G1	Closed	Module D constructs the completed tensor and skin targets, proves all eight target-relative fibres inhabited and final, and establishes one globally compatible bijective assignment.
G2	Closed in the frozen matching envelope	Module A supplies attachment-safe tensor-source descent; Module B proves common attached Noether/Casimir factorization.
G3	Closed in the frozen matching envelope	Module B proves one natural projection-wide factor and common calibrated equality.
G4	Closed in Module A scope	Exact Ward status and finite signed-incidence composite closure.
G5	Closed in Module C scope	Independently formed fixed-clock RPI histories, lawful projection descent, four-role intertwining, directed exchange, and strict/open cube coherence.
G6	Closed conditionally	TB1–TB11 remain explicit premises of the small-body/geodesic implication.
G7	Closed	Module D proves candidate, eligibility, compatibility, failure, live-residue, lawful-automorphism, and one-orbit closure, followed by Role-Occupancy licensure.

Contents

Read This First: Modular Proof Spine	2
I Proof Class, Target, and Kernel	9
1 Exact target and proof-class lock	9
2 Target adequacy forces the kernel	10
3 The fixed-domain consequence layer	11
4 Dependency inversion and no-backflow	12
5 Cost of kernel denial in the capstone target	13
6 Reportability and the fixed theorem boundary	13
II Theorem-Use Packages and Source Locks	14
7 Why theorem-use packages are necessary	14
8 Kernel and singleton-interior package	14

9	Realized physical interior package	15
10	AER gravity package	16
11	Einstein metric-projection package	16
12	Projective-Hilbert and Schrödinger package	17
13	Mass, inertia, and numerical-standing package	17
14	Accepted Module A source-representation package	18
15	Accepted Module B mass-charge package	19
16	Accepted Module C upstream-continuation package	20
17	Accepted Module D certificate-completion package	21
18	Record and compatibility packages	22
19	Coherent trajectory and transport packages	23
20	Role-occupancy closure package	23
21	Structural-unification package	24
22	The theorem-use closure ledger	24
III	The Unloaded Acyclic Pre-Certificate Target	26
23	The frozen burden is not placed inside the target	26
24	The eight forced certificate roles	27
25	Independent role interfaces	28
26	Global compatibility before occupation	30
IV	Typed Common Response, Constructed Source, and Composite Closure	32
27	The full response quotient is not compressed to one number	32
28	Certified composition is indexed by explicit cross-term data	33
29	The inertial transport quotient	34
30	Accepted Module A source admission and representation	35

31	Exact common-response domain after accepted Modules A and B	38
32	No independent same-interior scalar discriminator in the matching envelope	39
V	Accepted Module C: Upstream Continuation and Projection Intertwin- ing	41
33	Upstream formation before projection	41
34	Independently complete role fibres	42
35	Descended processes and projection functors	42
36	Distinct closed prediction and actual exchange continuation	43
37	The accepted continuation four-cube	44
VI	Complete Joint-Ledger Closure after Accepted Modules A–D	45
38	Ledger type	45
39	Accepted metric-source coordinate	45
40	Accepted quantum and common-response coordinates	46
41	Accepted record and compatibility coordinates	46
42	Accepted continuation and exchange coordinate	47
43	Certificate and report coordinates	47
44	Complete joint-ledger closure	48
VII	Accepted Module D: Complete Certificate Inhabitation	49
45	Canonical target diagram and eight accepted occupants	49
46	Coordinate inhabitation and target-relative finality	50
47	Eligibility, compatibility, and complete inhabitation	50
VIII	Accepted Module B Integration: Common Mass Charge and Inertial– Active Proportionality	52
48	Exact accepted matching envelope	52

49	Operator/source ancestry from accepted Module A	53
50	Global charge and stable mass Casimir	53
51	Active zero-transfer metric-source branch	54
52	Inertial Bargmann–Schrödinger branch	55
53	Common charge, naturality, and one factor	55
54	Transport extension and exact boundary	56
55	Passive role after accepted Module C	56
56	Violation taxonomy after Module B	57
IX	Declared Test-Body Scope and Weak-Equivalence Closure	58
57	Why the motion theorem is separate	58
58	The conventional Geroch–Jang theorem	58
59	AASC test-body certificate	58
60	Hypothesis-by-hypothesis occupation	59
61	Geodesic motion	60
62	Weak-equivalence closure	61
63	Relation to stronger small-body results	61
64	Nonclaims and boundary of the weak-equivalence result	62
X	Accepted Module D: Exact Fibre Exhaustion and Role-Occupancy Closure	63
65	Completed antecedent certificate	63
66	Candidate, quotient, compatibility, and failure completeness	63
67	Live residue and assignment space	64
68	Eight target-relative fibre theorems	64
69	Live-residue identification and role-realization completeness	65
70	Lawful automorphisms and unique orbit	65

XI	Adversarial Map for the Modules A–D Closure Release	66
71	Adversarial-map gate	66
72	Accepted Module A attacks	66
72.1	“Every upstream response is still declared source-bearing”	66
72.2	“The arbitrary zero functional still passes”	66
72.3	“The tensor distribution is merely notation”	66
72.4	“Ward closure is assumed”	67
72.5	“Composite closure hides interaction exchange”	67
73	Accepted Module B attacks	67
73.1	“The Casimir is still obtained from a generic state expectation”	67
73.2	“The operator EMT can still be unrelated to the accepted Module A source”	67
73.3	“Projective-state invariance still conflicts with quadratic homogeneity”	67
73.4	“The active monopole omits binding or stress content”	67
73.5	“Module B proves mass equality in every matter or spacetime regime”	68
74	Accepted Modules C and D attacks	68
74.1	“The manuscript constructs an upstream GR/QM continuation”	68
74.2	“The unique certificate orbit is inferred from role tags”	68
75	Same-scope rival normal forms	68
76	Attack-summary ledger	69
XII	Modules A–D Integration Theorems and Final Frozen-Endpoint Closure	70
77	Accepted Module A integration theorem	70
78	Accepted Module B integration theorem	71
79	Accepted Module C integration theorem	71
80	Accepted Module D integration theorem	72
81	Conditional weak-equivalence theorem	73
82	Final scope and reportability	73
83	Conclusion	74
A	Formal Predicates and Type Lock	75
A.1	Kernel-first target	75
A.2	Response and inertial types	75
A.3	Attachment-safe source-bearing type	76

A.4	Accepted source representation	76
A.5	Finite composite closure	77
A.6	Exact common-response overlap	77
A.7	Accepted Module C continuation type	77
A.8	Accepted Module D completion type	78
B	Certificate Coordinate and Final Integration-Status Ledger	79
C	Theorem-Use Provenance Ledger	80
D	Test-Body Hypothesis Map	82
D.1	Comparator routes	83
E	Adversarial Attack Map	83
F	Claim-Status and Reportability Matrix	84
G	Scope Boundary and Extension Program	86
G.1	Strong equivalence and self-gravity	86
G.2	Self-force, spin, and multipoles	86
G.3	Locally covariant quantum field source extraction	86
G.4	Nontrivial interaction dynamics	86
G.5	High-curvature and horizon endpoints	86
G.6	Parameter and numerical closure	87
H	Source Map and Dependency Index	87
I	Proof-Gate Closure Ledger	88
	References	90

Part I

Proof Class, Target, and Kernel

1 Exact target and proof-class lock

The paper's target is not “a theory that unifies GR and QM” in the usual equation-first sense. That phrase leaves the target under-typed. It can refer to a formal map between two mathematical languages, an effective interaction model, a quantum field theory on a background metric, a canonical quantization, a path integral, an emergent-spacetime proposal, a record-conditioned semiclassical model, or a complete physical realization. These objects have different carriers, admissibility envelopes, continuation laws, and reportable outputs. A proof cannot establish all of them by using the word *unification* as if it were a common type.

The fixed target here is narrower and structurally stronger. It is a single nondegenerate physical-interior theorem object whose standing-bearing roles admit a faithful gravitational constraint projection, a faithful closed projective-Hilbert projection, a standing-bearing record projection, a compatibility projection, and at least one nonidentity upstream continuation from which the corresponding downstream continuations are obtained. The target also requires an explicit certificate showing that all declared roles are occupied by eligible objects and that all role incidences are jointly compatible. The manuscript therefore studies a *complete certified realization of paired bookkeeping projections*, not a primitive map from one projection into the other.

Definition 1.1 (Capstone target). A *capstone target* on a fixed domain D is a tuple

$$\text{CapTarget}_D = (I_D, \mathcal{R}_D^{gg}, \mathcal{O}_D^{gg}, \text{Elig}_D, \text{Compat}_D, \text{Cont}_D, \text{Report}_D)$$

with the following intended burden:

1. I_D is one determinate realized physical interior;
2. $\mathcal{R}_D^{gg}$ is the finite set of required certificate roles;
3. $\mathcal{O}_D^{gg}$ is the set of candidate theorem-object occupants;
4. Elig_D fixes type-correct role incidence;
5. Compat_D fixes pairwise and global compatibility;
6. Cont_D is standing-preserving upstream continuation;
7. Report_D fixes the exact theorem status of the output.

The target is realized only by a complete eligible compatible role assignment with faithful metric, quantum, record, compatibility, transport, and matching projections.

Definition 1.2 (Correctness-relevant criticism). A criticism is correctness-relevant to this manuscript only if it identifies at least one of the following: a false target premise; failure of target adequacy to force the kernel; a broken imported theorem-use condition; an ineligible certificate occupant; a missing required role; a false compatibility edge; a failed continuation square; a hidden selector; a false common-descent factorization; an invalid uniqueness argument; an unmet test-body premise; or a report that exceeds the certified scope.

Remark 1.3 (Framework preference is not a counterexample). A demand that the proof be rewritten as canonical quantization, path integration, semiclassical gravity, string theory, loop quantum gravity, or a conventional equivalence-principle derivation changes proof carrier unless it identifies a failure in one of the named links. Such programs remain admissible richer-scope comparators. They are not the standing criterion for the present theorem.

2 Target adequacy forces the kernel

AASC is not an optional interpretive overlay on independently complete physical content. The target itself requires the kernel. To use a physical interior, a certificate role, or a continuation as a determinate theorem object, the proof must already distinguish licensed from unlicensed use, preserve the same target across redescription, determine whether an output may bear downstream force, and retain the difference between failure and later repair.

Definition 2.1 (Kernel-neutral target adequacy). A theorem-bearing physical-interior target is *target-adequate* when it satisfies:

1. **Target determinacy:** the same object of evaluation is fixed across use and lawful redescription;
2. **Status evaluability:** assertions, transitions, and certificate incidences can stand, fail, or remain unlicensed relative to that target;
3. **Act-time finality:** a later correction or enriched certificate is a new act, not retroactive success of the original failed act;
4. **Same-regime fidelity:** lawful redescription preserves target, carrier, role, standing status, and continuation profile.

Theorem 2.2 (Target adequacy forces the kernel). *If CapTarget_D is target-adequate and nondegenerate, then the kernel*

$$K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{lrr}_D)$$

is already in force for the target.

Proof. Target determinacy requires a role preserving which object is evaluated through reuse and redescription; this is reference. Status evaluability requires a distinction between an inscription and an output with licensed downstream force; this is standing. Act-time finality requires that a later repair not change the status of the earlier act while it remains the same act; this is irreversibility. Same-regime fidelity requires a stable boundary separating licensed continuation, declared enrichment, bookkeeping, support, repair, and failure; this is admissibility. None of these roles is introduced from the later metric or Hilbert projection. They are the governance work already performed by the target as a determinate theorem object. $\square$

Corollary 2.3 (Vocabulary independence). *Rejecting the names **Adm**, **Stand**, **Ref**, and **Irr** does not remove the kernel if the same governance work is retained. A genuine denial must preserve the capstone target while omitting one role and without reintroducing its work under another name.*

Proposition 2.4 (No third position). *There is no position that rejects the kernel while retaining the same determinate complete-certificate target. Either the target remains nondegenerate, in which case definition 2.2 applies, or target identity, theorem-bearing status, act-time finality, or same-regime fidelity is lost.*

3 The fixed-domain consequence layer

The paper uses a compact consequence package, denoted A_D^+ . It is not an additional ontology. It records the fixed-domain consequences of the kernel that are required to type certificate assembly.

Definition 3.1 (Capstone consequence layer). For the fixed target, A_D^+ contains:

1. **AMetric nonselection**: no metric, Hilbert norm, probability, scalar mass, coupling, basis, branch, coordinate, scale, or representative has primitive boundary authority;
2. **No second same-scope classifier**: a factor changing standing is governance-equivalent, declared anchor motion, declared tensor load, or forbidden hidden selector work; if it changes no standing, it is skin;
3. **No same-act repair**: later source completion, collapse, record insertion, renormalization choice, or certificate addition cannot repair an earlier failure as the same act;
4. **Standing as reusable admissibility**: a standing output is one whose admissibility survives licensed continuation;
5. **Singleton interior**: same-domain plural admissible interiors collapse to lawful equivalence or reveal a competing gate, scope split, or standing failure;
6. **Anchor/tensor/skin trichotomy**: every descriptive element is domain-fixing anchor, standing-bearing tensor/operator/compatibility content, or presentation skin;
7. **No projection backflow**: a downstream metric or Hilbert fact cannot authorize the interior or another projection;
8. **No undeclared carrier transfer**: a change of dimension, signature, connection, field content, state carrier, record regime, boundary, or continuation law changes scope unless declared;
9. **Report preservation**: no report may claim more theorem force than the certificate bears.

Theorem 3.2 (No independent certificate classifier). *Within the fixed capstone target, no independent same-scope certificate classifier can carry theorem force outside the kernel-governed eligibility and compatibility interfaces.*

Proof. Suppose a classifier F changes whether an occupant may fill a role or whether a completed assignment may stand. Then F performs admissibility-relevant work. If F preserves the kernel roles and produces exactly the existing eligibility or compatibility decision, it is governance-equivalent

bookkeeping. If it changes domain, carrier, boundary, continuation, or theorem type, it is declared anchor motion. If it contributes a new standing-bearing witness, it is declared tensor or operator load and the target is enriched. If it changes no downstream use, it is skin. The remaining case is undeclared standing authority, which contradicts the fixed admissibility boundary. $\square$

4 Dependency inversion and no-backflow

The central error this paper must avoid is the following inversion:

$$\text{Einstein equation} + \text{Schrödinger equation} \implies \text{physical unity.}$$

That expression treats the projections as independently complete and asks their juxtaposition or interaction to create unity. The kernel order is instead

$$\text{determinate physical target} \implies K_D \implies I_D \implies (\Pi_g(I_D), \Pi_H(I_D)).$$

The two equations may confirm that their projections are faithfully occupied; they do not create the object projected.

Definition 4.1 (Projection backflow). Projection backflow occurs when a downstream object is used to authorize, generate, select, repair, or reinterpret the upstream standing structure that licensed its projection, or when one projection is used to authorize another without a declared common carrier and joint ledger.

Theorem 4.2 (Dependency-inversion exclusion). *No metric equation, Hilbert-space fact, Hamiltonian operator, stress-energy expression, mass scalar, test-body theorem, or empirical success can retro-authorize K_D , A_D^+ , or I_D within the same proof act.*

Proof. Each listed object is formulated only after a carrier, target, redescription law, continuation rule, and reportable status have been fixed. Using the object to authorize those prior conditions makes a downstream tensor or representative into a boundary selector. If the added authority changes the domain or tensor ledger, the proof has changed scope. If it changes no standing-bearing content, it is skin. If it changes standing without declaration, it violates definition 3.2. If it attempts to repair a previously unlicensed target, it violates irreversibility. Hence no same-act backflow survives. $\square$

Corollary 4.3 (The test-body theorem is downstream). *The geodesic theorem used in Part IX is a metric-projection theorem. It cannot prove the kernel, the physical interior, the gravity role, or the common deformation-response role. It may be applied only after those objects and the metric projection are licensed.*

5 Cost of kernel denial in the capstone target

Denied role	Capstone-specific cost
Reference	No fixed physical interior, required-role set, occupant identity, projection map, continuation square, or common source class.
Standing	Certificate entries remain annotations or comparisons without licensed theorem force or downstream reuse.
Admissibility	No boundary separates eligible occupation from hidden load, declared enrichment, support, calibration, repair, or report inflation.
Irreversibility	Missing roles, failed squares, nonconserved sources, or unfixed records can be patched later and laundered into the original certificate.

The table is not rhetorical. Each cost is a direct loss of the theorem object. A critic may reject the target and pursue a different program. That is a legitimate scope choice. It is not a same-target counterexample.

6 Reportability and the fixed theorem boundary

Definition 6.1 (Reportable capstone output). A reportable capstone output is a theorem-bearing statement whose content is exactly the complete certificate class, its proven factorization and commutation properties, and the explicit scope of its conditional test-body consequence.

Proposition 6.2 (No all-dynamics promotion). *The complete certificate theorem does not imply uniqueness of all richer-scope quantum-gravity dynamics, a new nonzero interaction law, all curved-spacetime QFT, or the strong equivalence principle.*

Proof. Those targets add carrier, field, boundary, analytic, interaction, or self-gravity burdens not included in CapTarget_D . By the no undeclared carrier-transfer consequence, their addition changes the theorem object. The present uniqueness result is therefore unique complete role assignment for the fixed certificate, not uniqueness of every possible enriched dynamics. $\square$

Part II

Theorem-Use Packages and Source Locks

7 Why theorem-use packages are necessary

The present manuscript is a capstone composition theorem. It would be poor mathematical practice to reproduce the full kernel exhaustion, RPI ladder, AER coisotropic proof, Einstein curvature-response exhaustion, Wigner–Stone bridge, Hamiltonian principal-symbol proof, record-fixation theorem, and Bell-scope compatibility theorem in one document. Repetition would obscure the genuinely new burden and create multiple independently editable copies of the same theorem. The correct method is to import each source only at a fixed theorem-use interface and to state the hypotheses, conclusion, permitted use, and prohibited promotion.

Definition 7.1 (Theorem-use package). A theorem-use package is a tuple

$$\text{TU} = (\text{Source}, \text{Hyp}, \text{Conclusion}, \text{Use}, \text{Boundary})$$

whose source theorem may be invoked in this manuscript only under **Hyp**, only for **Conclusion**, only at **Use**, and never beyond **Boundary**.

Proposition 7.2 (No source laundering). *A theorem-use package may not be used to promote a source's support, notation, illustration, conditional corollary, or future program into a premise or conclusion of the capstone theorem.*

Proof. Such promotion changes the imported theorem object. If the promoted material adds standing-bearing content, it is undeclared tensor or anchor load. If it changes no standing-bearing use, it is skin. If it is used to repair a missing capstone role, it is source laundering and violates no same-act repair. Therefore only the declared theorem-use conclusion is admissible. $\square$

8 Kernel and singleton-interior package

The kernel source proves that nondegenerate determinate same-domain construction already requires admissibility, standing, reference, and irreversibility [9]. The singleton-interior source closes plural same-domain admissible interiors and competing gate structures [10]. The present paper uses these results only to fix the one-interior target and exclude a gravity-side/quantum-side split.

Theorem 8.1 (Kernel theorem-use package). *For the capstone target,*

$$\text{TargetAdeq}_D \implies K_D \implies A_D^+.$$

The permitted use is target governance, fixed-domain role typing, no second classifier, no same-act repair, no projection backflow, and report preservation. The package supplies no metric, Hilbert space, field content, source tensor, Hamiltonian, mass value, record, or test-body theorem.

Theorem 8.2 (Singleton-interior theorem-use package). *Let I_D be the admissible physical interior fixed by one target, admissibility boundary, standing classification, reference structure, and continuation profile. Any purported second same-domain admissible interior is lawfully equivalent to I_D , skin-level duplication, competing-gate import, scope change, hidden carrier/repair, or standing failure.*

Corollary 8.3 (No gravity/quantum interior split). *There do not exist distinct same-domain admissible interiors I_g and I_H such that gravity descends from I_g , quantum standing descends from I_H , and the two are merely coordinated externally while preserving the same capstone target. The admissible form is*

$$I_D \implies (\Pi_g(I_D), \Pi_H(I_D)).$$

9 Realized physical interior package

The RPI theorem constructs the minimal role architecture required for standing-bearing physical persistence and proves coherence, closure, and operational minimality [11]. It is important to state its exact function. RPI is not a lower-level material generator. It is the necessity-conditioned physical specialization of the kernel. Its role sequence includes non-null persistence, bounded localization, continuation domains, quotient-stable comparison, bounded transport, fields, particles, gauge, mass/inertia, vacuum, time, and causality.

Theorem 9.1 (RPI theorem-use package). *For a fixed admissible realized persistence regime D ,*

$$K_D \wedge \text{FixedAdmRealPers}_D \wedge \text{RoleSeq}_D \implies I_D \wedge \text{Coher}_D(I_D) \wedge \text{Closure}_D(I_D) \wedge \text{Minimal}_D(I_D).$$

The role sequence contains

$$\begin{aligned} &\text{NonVacRec}_D, [B_D]_{\equiv_B}, (\text{Adj}_D, N_D, \text{LCD}_D), \text{Cmp}_D, \text{BoundTrans}_D, \\ &\Phi_D, \Pi_D, G_D, M_D = (\mu_D, \iota_D), \Omega_D, T_D, \text{Cau}_D. \end{aligned}$$

The permitted use is to supply the one coherent role architecture over which the capstone certificate is assembled. The package does not by itself supply a complete paired-projection certificate, an explicit role-occupancy assignment, a unique certificate orbit, or a test-body theorem.

Remark 9.2 (RPI is input, not the capstone conclusion). The distinction between RPI and the present paper is exact. RPI proves that the roles must form one coherent, closed, minimal physical interior. This paper defines the certificate roles required by a specific paired-projection realization, supplies an explicit occupant for each role, proves global compatibility and projection commutation, and computes the lawful assignment quotient. Repeating only $K_D \Rightarrow I_D$ would be redundant; executing the certificate over I_D is not.

10 AER gravity package

The gravity theorem identifies gravity as the constraint-governance role forced when the admissible physical interior is specialized to regular-stratum observable Hamiltonian evaluation [12, 13]. The proof begins with a standing surface C in a symplectic carrier, a quotient map $\pi : C \rightarrow S_{\text{phys}}$, and the redescription distribution $D = \ker(d\pi)$. Under extension-independent Poisson descent, quotient separation, and observable maximality, admissible redescription is identified with the characteristic distribution of the pulled-back symplectic form. Coisotropic normal form then yields local constraints, first-class closure, and multiplier presentation.

Theorem 10.1 (AER/gravity theorem-use package). *In the regular-stratum AER comparison class,*

$$I_D + \text{AER}_{\text{Obsmin}} + \text{AER}_{\text{Sep}} + \text{AER}_{\text{Max}} + \text{AER}_P \implies \text{Gravity}_D,$$

where Gravity_D consists, up to lawful role equivalence, of:

1. a coisotropic standing surface;
2. physical observables factoring through a quotient;
3. admissible redescription equal to the characteristic foliation;
4. local first-class constraint generators;
5. multiplier-type presentation dynamics.

The package supplies the gravity role before metric dynamics. It does not supply the Einstein equation, numerical gravitational coupling, all singular strata, all boundary completions, or a quantum-gravity interaction law.

Corollary 10.2 (Gravity precedes metric projection). *The metric, curvature tensor, Einstein tensor, and source equation may represent the gravity role only after the role has been fixed. They cannot be premises for definition 10.1.*

11 Einstein metric-projection package

The Einstein source proves a target-specific normal form inside a faithful four-dimensional Lorentzian metric projection [14]. The same-scope first-response predicate permits only metric-internal, local, symmetric rank-two response linear in the first curvature obstruction, with no undeclared extra field, connection, nonlocal, boundary, or source-side load. The geometric candidates reduce to $R_{\mu\nu}$, $Rg_{\mu\nu}$, and $g_{\mu\nu}$. Closed source continuation and the contracted Bianchi identity fix the Einstein combination.

Theorem 11.1 (Einstein theorem-use package). *Let $\Pi_g(I_D) = (M^4, g, \nabla)$ be a faithful Lorentzian metric projection with minimal metric compatibility. If $E_{\mu\nu}$ satisfies $\text{SameScopeMetric}_1(E)$ and $T_{\mu\nu}^{\text{closed}}$ is the closed metric-facing source ledger, then*

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu},$$

closed source continuation implies $\nabla^\mu E_{\mu\nu} = 0$, and the nontrivial curvature branch yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

The source tensor is the metric-facing extraction of already-admitted field-carrier, particle-persistence, deformation-response, inertia/transport, interaction/restoration, and vacuum-baseline roles.

Nonclaim 11.2 (No primitive source substance). The theorem-use package does not identify $T_{\mu\nu}$ with a primitive mass-energy substance. Its capstone use is precisely as a typed extraction from the upstream source ledger.

Nonclaim 11.3 (No numerical constants). The package fixes equation form. It does not generate numerical values of κ , Λ , or matter-sector masses. Those require scale and calibration anchors.

12 Projective-Hilbert and Schrödinger package

The quantum theorem first fixes ray/projective state standing and transition-probability geometry. Wigner implementation, continuous branch selection, and Stone generation produce closed unitary continuation with a self-adjoint generator. The Schrödinger source then exhausts the minimal scalar nonrelativistic local Hamiltonian response [15, 16, 17, 18]. Under the fixed spatial envelope and no hidden gauge, spin, relativistic, nonlocal, open, update, or boundary load, the generator is $a\Delta + M_V$. The coefficient is fixed after the inertial/action anchors are admitted.

Theorem 12.1 (Quantum/Schrödinger theorem-use package). *Let $\Pi_H(I_D)$ be a faithful closed scalar nonrelativistic projective-Hilbert projection on the declared domain. Transition-standing-preserving reversible continuation gives a strongly continuous unitary group and self-adjoint generator H . If H satisfies SameScopeQuantumSpatial₁(H), then*

$$H = a\Delta + M_V.$$

After admitted action and inertial scale anchoring,

$$H = -\frac{\hbar^2}{2m}\Delta + M_V, \quad i\hbar\partial_t\psi = H\psi.$$

The coefficient m represents the licensed inertial/transport response of the localized persistence class; it is not a primitive scalar source.

Nonclaim 12.2 (No all-quantum promotion). The package does not cover spin, gauge connection, relativistic wave equations, open-system dynamics, measurement update, nonlinear effective equations, QFT, or quantum-gravity dynamics except as declared richer scopes.

13 Mass, inertia, and numerical-standing package

The mass source isolates the upstream deformation-response profile required for a localized particle persistence class to remain the same through admissible deformation and transport change [19]. MGN1 then establishes the standing quotient and blocks numerical proliferation, coordinate inflation, normalization selectors, empirical-fit selectors, and witness-free promotion [20].

Theorem 13.1 (Mass/inertia theorem-use package). *For every admitted particle persistence class Π ,*

$$\mu_D(\Pi) : \Delta_\Pi \rightarrow C_D, \quad \iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{\text{tr}}}.$$

Here μ_D is the quotient-stable standing-preserving compression/deformation-response profile and ι_D is its transport-facing restriction. Numerical mass parameters are later representations of this role when their use is witness-bearing, quotient-stable, gauge compatible, and scale disciplined.

Theorem 13.2 (MGN1 theorem-use package). *A deformation-associated numerical occurrence has standing only if it is attached to an admitted deformation witness, stable under the inherited standing quotient, compatible with bounded transport and the fixed survivor ledger, residual/vacuum compatible, non-selector-generated, and irreducible to unit, normalization, basis, representative, regulator, scheme, or coordinate skin. A downstream embedding does not promote witness-free numerical content unless it supplies a new admitted witness that changes the quotient.*

Corollary 13.3 (No bare gravitational-mass coordinate). *A scalar labeled “gravitational mass” cannot acquire same-interior source standing merely by being inserted into $T_{\mu\nu}$ or compared with the Hamiltonian parameter m . It must factor through the admitted deformation/source quotient or be typed as calibration, declared tensor load, scope change, skin, or selector.*

14 Accepted Module A source-representation package

The repaired source theorem object is imported from *Operational Metric-Source Representation and Composite Ledger Closure under AASC*, module MASRC_{1.0}, release v1.1.0-attachment-corrected [1]. Its source-admission, representation, Ward, composite, and software coordinates were all judged correct in the bounded independent re-review [2]. The source manuscript and positive verdict are embedded in this project under `modules/ModuleA_MA-SRC-1.0_v1.1.0/` and `docs/moduleA_review/`.

Theorem 14.1 (Module A theorem-use package). *Let Q_{resp} be the upstream response quotient fixed before metric source representation. Module A supplies:*

1. *an independently formed attachment relation $\text{Attach}_g(p, \mathfrak{a}_p, \mathcal{W}_p)$ requiring the graph of $\mathcal{W}_p$ to equal the metric-response relation generated by the complete upstream response protocols and the tangent of the already licensed faithful metric projection;*
2. *the genuine base subtype $Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$ and the structured quotient $\hat{Q}_{\text{src}}$, with a surjective forgetful map $F : \hat{Q}_{\text{src}} \rightarrow Q_{\text{resp}}^{\text{src}}$;*
3. *a unique natural symmetric tensor-density distribution representing every complete attached metric witness and the quotient-faithful map*

$$q_g : \hat{Q}_{\text{src}} \longrightarrow \text{Src}_g^{\text{dist}} / \simeq_{\text{law}};$$

4. *the exact distributional Ward-divergence equivalence*

$$\mathcal{W}_x(\mathcal{L}_X g) = -\langle J_x, X \rangle \iff \nabla_\mu \mathbf{T}_{x\nu}^\mu = J_{x,\nu};$$

5. *finite completed-composite closure by exact term coverage, nonduplication, signed internal-exchange balance, declared external currents, and boundary/regularity certificates;*

6. a bounded software correspondence checking only the finite registry and incidence conditions.

The package does not prove that every response is source-bearing, does not identify the base subtype with the structured quotient, does not construct a universal active-mass scalar, does not prove inertial–gravitational proportionality, does not construct an upstream gravity–quantum continuation, and does not prove a unique certificate orbit.

Corollary 14.2 (Accepted source and composite gates). *The metric-source existence/descent branch of $G2$ and the complete finite composite-source gate $G4$ may be imported from definition 14.1 provided the capstone preserves the exact source-bearing types and does not promote Module A beyond its frozen boundary.*

15 Accepted Module B mass-charge package

The repaired mass-charge theorem object is imported from *Common Mass-Charge Matching and Universal Inertial–Active Proportionality under AASC*, module MBMASS_{1.0}, release v1.3.0-normalized-ray-source-lift-corrected with the authorized Cycle-4 four-command typesetting patch [3]. The bounded independent Cycle-4 review judged the frozen mathematical endpoint correct and authorized import after the diagram-only correction [4]. The complete patched project is embedded under modules/ModuleB_MB-MASS-1.0_v1.3.0_diagram-patched/; the positive report and machine-readable verdict are embedded under docs/moduleB_review/.

Theorem 15.1 (Module B theorem-use package). *Let $\hat{Q}_{\text{mass}}^{\text{op}}$ be the operator-presentable refinement of the attachment-certified mass-bearing overlap, restricted to the frozen stable positive-mass spin-zero Poincaré–Noether/form-factor/Breit/Bargmann matching envelope. Module B supplies:*

1. *a normalized-ray family of concrete accepted Module A source presentations formed before lawful quotienting, its norm-squared homogeneous vector extension, and the unique Hermitian operator-valued total energy–momentum distribution $\hat{\mathbf{T}}_x$ reconstructed by complex polarization;*
2. *inheritance by $\hat{\mathbf{T}}_x$ of the accepted source support, naturality, Ward, boundary, improvement, and integrated-charge ledgers;*
3. *equality of the global charge of $\hat{\mathbf{T}}_x$ with the Stone translation generator on the full irreducible reference representation, followed only afterward by restriction to the stable scalar mass sector;*
4. *exact total-EMT zero-transfer normalization $A_x(0) = 1$, a regulated stationary Breit transition source, exact Laue closure, and the active linearized Einstein monopole*

$$h_{00}(r) = \frac{\kappa c^2 m_{\text{Cas}}(x)}{4\pi r} + O(r^{-2});$$

5. *an explicit common-core contraction of the massive scalar Wigner representation to the Bargmann/Galilei regime, yielding central mass $m_{\text{Cas}}(x)I$ and*

$$H_{\text{kin}} = -\frac{\hbar^2}{2m_{\text{Cas}}(x)}\Delta;$$

6. *one common intrinsic Noether/Casimir charge line through which the active zero-transfer source scalar and inertial Hamiltonian scalar factor while retaining their distinct codomains;*
7. *one projection-wide factor*

$$m_g(x) = \alpha_D m_i(x), \quad \alpha_D = \frac{\kappa c^2}{4\pi\beta_g} > 0,$$

for every admitted stable scalar species and completed stable scalar composite in the fixed matching envelope;

8. *calibrated equality $m_g = m_i = m_{\text{Cas}}$ after the common Einstein–Newton/Bargmann calibration, plus naturality and flat calibration-transport extension on the declared connected envelope.*

The package does not identify m_{Cas} with the norm of a generic state’s mean four-momentum or with an arbitrary localized-profile energy. It does not itself supply a generic conserved energy on curved spacetime, a passive/free-fall continuation, spin or multipole closure, unstable or radiating sectors, upstream history construction, Module D fibre exhaustion, strong equivalence, or numerical mass prediction.

Corollary 15.2 (Accepted common-response and proportionality gates). *Within the exact frozen Module B matching envelope, the remaining common-charge branch of G2 and the universal source-naturality/proportionality gate G3 may be imported from definition 15.1. The import does not close G1, G5, G7, or any claim outside the stable scalar source-attached matching subtype.*

16 Accepted Module C upstream-continuation package

The repaired continuation theorem object is imported from *Kernel-Native Upstream Continuation and Projection Intertwining under AASC*, module `MCCONT1.0`, release v1.2.1-editorial-clarified-import-authorized [5]. The bounded independent Cycle-3 review judged the unchanged standalone mathematical endpoint correct and authorized capstone import [6]. The complete clarified project is embedded under `modules/ModuleC_MC-CONT-1.0_v1.2.1_editorial-clarified/`; the positive report, verdict, exact editorial diff, and rendered clarification pages are embedded under `docs/moduleC_review/`.

Theorem 16.1 (Module C theorem-use package). *Within the declared regular local fixed-clock envelope, Module C supplies:*

1. *one certified local clock representative and a smooth, locally Lipschitz, clock-normalized RPI-level seed formed before any metric, Hilbert, Hamiltonian, or solved downstream history is evaluated;*
2. *the fixed-clock Moore-history category $\mathbf{Cont}_D^\tau$, with strict concatenation, identities, standing preservation, fail-closed prefixes, and no deferred repair;*
3. *orientation-preserving clock changes as duration-transport equivalences between separately constructed presentation categories rather than automorphisms of one fixed Moore category;*
4. *independently complete metric/source, quantum, record, and compatibility fibres, with the metric/source object given by a lawful groupoid orbit, the quantum object fixed by the complete*

transition/process/generator-domain/spatial-source profile, and the latter two by their final imported quotients;

5. *unique descended role processes: smooth projectability for regular metric and quantum profiles, and exact quotient preservation for record and compatibility profiles;*
6. *on the exact Cauchy-eligible subtype, intertwining of the metric image with the independently fixed Einstein–Module-A development;*
7. *on the exact full-ray quantum-eligible subtype, a strongly continuous transition-preserving process classified downstream by Wigner, Stone, and the accepted Schrödinger normal form;*
8. *record-lineage and compatibility-not-transport descent from the same upstream history;*
9. *distinct closed-prediction maps $(\Pi_X^0, \widehat{\beta}_X^0)$ and actual exchange-extended maps $(\widetilde{\Pi}_X, \widetilde{\beta}_X)$, linked by directed, generally noninvertible exchange cells generated by one common incidence monoid;*
10. *the source-target-correct exchange cocycle and a genuine sixteen-vertex continuation four-cube whose faces and monotone paths commute by whiskered coordinatewise pasting; zero incidence gives the strict closed cube;*
11. *one complete closed nonidentity RPI packet and one complete declared-open fixed-background packet inhabiting every mandatory Module C coordinate;*
12. *a bounded software checker whose claims are restricted to the finite registry, incidence, endpoint, vertex, and cocycle conditions actually implemented.*

The abstract full-ray carrier in the complete packet uses its standard smooth projective-Hilbert-manifold structure within the regular local RPI chart only to certify chart regularity. That representative smooth structure has no upstream authority over the kernel, the realized physical interior, or the continuation seed.

The package does not claim a global preferred clock, global completeness of every RPI flow, nonzero interacting quantum gravity, arbitrary distributional Einstein Cauchy well-posedness, that every quantum source is metric-source-bearing, Module D fibre uniqueness, the capstone assignment orbit, or strong equivalence.

Corollary 16.2 (Accepted continuation and intertwining gate). *G5 is closed in the exact frozen Module C envelope. The transport/continuation and full matching candidate coordinates are inhabited by accepted theorem objects. Final certificate closure is supplied only by the later accepted Module D package.*

17 Accepted Module D certificate-completion package

Module D is the final target-relative certificate-completeness theorem object [7, 8]. It does not create a source, state, seed, dynamics, clock, or interaction. It receives the exact accepted Modules A–C target instance and proves that its certificate-generated candidate response is complete, lawfully quotiented, globally compatible, and uniquely occupied.

Theorem 17.1 (Module D theorem-use package). *Relative to the exact frozen Modules A–C target, accepted Module D supplies:*

1. fifteen source-locked primary tensor generators and twenty-four primary overlap relations, completed covariantly by explicit left Kan path terms in

$$\text{TenInt}_D = \mathbf{Set}^{S_D}/\mathcal{W}_D;$$

2. exact forward-cosieve support closure and total same-direction incidence maps, with completed overlaps represented by their joint images in the adjacent generator fibre products;
3. the total tensor ledger

$$L_D^0 \cong \text{colim } \mathbb{G}_D$$

with its universal mediator property;

4. extension of the accepted probes by adjunction, an induced tensor probe, the complete joint probe, and the maximal lawful skin relation

$$S_D^0 = \text{KerPair}(\text{Probe}_{\Theta_D^{ABC}});$$

5. candidate, quotient, role-set, eligibility, compatibility, live-residue, and failure completeness;
6. eight separate target-relative fibre-finality theorems for anchor, tensor, skin, GR/source, quantum, boundary, transport, and matching roles;
7. one canonical compatible bijection from the forced role set onto the independently formed live residue;
8. one lawful role-incidence assignment orbit,

$$\left| \Sigma_D^{qq}/\Gamma_D^{\text{lawful}} \right| = 1;$$

9. role-realization completeness and certificate-relative realization licensure for the exact frozen target.

The package does not assert uniqueness across different source, potential, state, initial-data, seed, or matching target instances and does not add new physical dynamics.

Corollary 17.2 (Accepted certificate and occupancy gates). *G1 and G7 are closed relative to the exact accepted Modules A–D target. The core complete-certificate realization endpoint of FEP-1.0 is therefore closed. G6 remains a separate conditional downstream theorem under TB1–TB11.*

18 Record and compatibility packages

Measurement records and entanglement enter only where their roles are required. The record source proves that theory-level retained and evidential record use factors through a fixed-domain record quotient and requires admissible fixation [21]. The entanglement source proves that Bell-scope joint content is boundary-level compatibility, not hidden metric transport or same-act repair [22, 23].

Theorem 18.1 (Record theorem-use package). *Any measurement, branch, collapse, decoherence, outcome, or observer-relative update used as metric-facing source content or reportable evidence must*

be carried by an admissible record ledger fixing record role, quotient identity, fixation commutation, lineage, downstream reuse, and reportability. Without that ledger, the use is record laundering, hidden update, open/effective scope, or standing failure.

Theorem 18.2 (Compatibility theorem-use package). *Entanglement and Bell correlations may carry tensor-level compatibility on a joint standing class. They do not by themselves carry metric transport, source transfer, communication, branch selection, horizon rescue, or standing repair. Any such additional role must be declared in the cross-response, transport, boundary, or richer-scope ledger.*

19 Coherent trajectory and transport packages

The coherent-trajectory source distinguishes a licensed transition from a standing-preserving trajectory and proves composition closure, invariance under identity-preserving redescription, and no deferred coherence [24]. The connection and covariant-differentiation sources type connection data as standing-preserving transport geometry and covariant differentiation as the projection-facing comparison of tensor content under licensed transport [25, 26].

Theorem 19.1 (Trajectory theorem-use package). *A legal trajectory is coherent iff every retained stage has standing. Coherent trajectories compose. Identity-preserving redescription preserves coherence. A failed prefix cannot be repaired by later extension as the same lineage.*

Theorem 19.2 (Transport theorem-use package). *Standing-bearing transport is defined on fixed roles and quotient identity. In a licensed metric projection, covariant differentiation records the change of tensor content relative to the admitted connection. A connection, derivative, boundary condition, or scale-transport map has standing only when its anchor and tensor roles are declared and continuation identity is preserved.*

20 Role-occupancy closure package

Role-occupancy closure distinguishes candidate residue from realized role occupation [27]. Bare cardinality equality is not sufficient; a complete assignment must cover all forced roles, satisfy pointwise eligibility, obey joint compatibility, and be unique modulo certificate-lawful automorphism.

Theorem 20.1 (Role-occupancy theorem-use package). *Let C be a fixed certificate and T a target with target-forced role set $\text{RoleSet}(C, T)$. Let $\text{Residue}_T^{\text{live}}(C, T)$ be the union of residues occurring in injective, role-covering, pointwise eligible, jointly compatible pre-assignments. Let $\text{Assign}(C, T)$ be the set of compatible bijections from the forced role set onto that live target-relative residue. If C is role-realization complete—candidate, quotient, role-set, eligibility, compatibility, live-residue, and failure complete—then*

$$\text{RealizationLicensed}(C, T) \iff \left| \text{Assign}(C, T) / \Gamma^{\text{lawful}}(C, T) \right| = 1.$$

The theorem supplies no generator for occupants and does not apply to arbitrary total maps. It licenses the unique compatible bijective occupation of an independently formed live residue under a complete certificate.

Remark 20.2 (How the capstone uses role occupancy). The present paper does not infer physical existence from the number one. It first fixes the eight roles, constructs role-specific candidate fibres, forms the presentation quotient and live residue, proves certificate completeness, proves that the canonical assignment is a compatible bijection onto the live residue, and only then applies definition 20.1. The final orbit is never used as an antecedent occupant.

21 Structural-unification package

The existing structural-unification manuscript proves singleton-interior paired projection, no projection primacy, joint-ledger necessity, compatibility-square failure classification, and cross-projection route exhaustion [28]. Its endpoint is necessity and classification. The capstone uses that theorem as the specification of what must be populated and then proves inhabitation for the constructed certificate.

Theorem 21.1 (Structural-unification theorem-use package). *For the singleton interior, Einstein and Schrödinger theorem objects are paired downstream projections. Neither projection authorizes the other. Any standing-bearing cross-projection use satisfying the fixed same-scope joint predicate is carried by a declared ledger*

$$\mathcal{L}_D^{gq} = (L_g, L_H, L_{\text{cross}}, L_{\text{rec}}, L_{\text{comp}}, C_{\text{cert}}, \text{Cont}_{gq}, \text{Report}).$$

Failure of the predicate is typed as projection backflow, source laundering, hidden update, record laundering, compatibility laundering, certificate failure, open/effective scope, measurement/update scope, richer scope, skin, or standing failure.

Corollary 21.2 (The capstone burden). *Accepted Modules A–C construct the ledger and its projection/continuation incidences. Accepted Module D proves exact fibre completion, compatible bijective occupation of the independently formed live residue, and one lawful assignment orbit. Parts III–X integrate those theorem objects into the final certificate-relative capstone conclusion.*

22 The theorem-use closure ledger

Theorem 22.1 (Import closure). *The theorem-use packages definitions 8.1, 8.2, 9.1, 10.1, 11.1, 12.1, 13.1, 13.2, 14.1, 15.1, 16.1, 17.1, 18.1, 18.2, 19.1, 19.2, 20.1 and 21.1 supply every imported role used by the capstone construction. No unlisted theorem object is allowed to carry load in the main proof.*

Proof. The required imported roles are: target governance; unique physical interior; coherent role architecture; gravity constraint role; metric response normal form; quantum continuation normal form; deformation-response source; numerical quotient discipline; attachment-safe metric-source representation and composite closure; common mass-charge matching and universal inertial-active proportionality in the fixed stable scalar envelope; upstream continuation and four-role intertwining; exact certificate-fibre exhaustion, live-residue closure, and unique occupation; record fixation; compatibility typing; trajectory coherence; transport; role-occupancy licensure; and joint-ledger specification. Each appears in exactly one or more declared packages above. The machine-readable

theorem-use ledger in the project records source, hypothesis, imported conclusion, use locus, and nonpromotion boundary. Therefore every imported premise is visible and typed. $\square$

Part III

The Unloaded Acyclic Pre-Certificate Target

23 The frozen burden is not placed inside the target

The second bounded hostile review correctly identified a target-loading regression in the previous release. A theorem cannot prove source-functional existence, composite closure, source order, or matched continuation by including those objects in the definition of the target. The present construction therefore separates three levels:

upstream target data $\longrightarrow$ objects constructed in this manuscript
 $\longrightarrow$ complete certificate and occupation closure.

Only the first level is antecedent. The second and third levels are theorem conclusions.

Definition 23.1 (Unloaded fixed capstone target). The fixed capstone target T_D^{gq} consists of the following already-established theorem-use data:

1. the singleton kernel-governed realized physical interior I_D ;
2. the AER gravity role and the declared first-response Lorentzian metric projection type;
3. the projective-Hilbert quantum-standing role and the declared closed scalar nonrelativistic Hamiltonian projection type;
4. the upstream mass/deformation-response role (μ_D, ι_D) , without a primitive scalar identification between its downstream representations;
5. the record-fixation and compatibility-not-transport theorem objects;
6. the coherent-trajectory, bounded-transport, and no-repair theorem-use packages;
7. the eight-coordinate certificate grammar of definition 24.1;
8. the frozen report boundary and nonclaims of FEP-1.0.

The target does *not* assume a metric source functional, composite source closure, an active-source order, a nonidentity matched continuation, a commuting projection cube, a live residue, a unique orbit, realization licensure, or weak equivalence. Each is a downstream proof obligation.

Definition 23.2 (Genuine realization anchors). The realization envelope is

$$\mathcal{E}_D = (D, \text{Scope}, \text{Reg}, \text{Boundary}, \text{Domains}, \text{Scales}, \text{Failure}, \text{Report}).$$

It fixes only genuine scope data:

1. a regular AER stratum and a four-dimensional time-oriented Lorentzian first-response metric envelope;
2. a globally hyperbolic local Cauchy patch, a declared differentiability/Sobolev class, and declared boundary behavior;
3. a closed scalar nonrelativistic projective-Hilbert envelope with a declared dense self-adjoint domain;
4. record and compatibility domains;
5. action, inertial, and metric-source calibration anchors, without fixing their numerical values;
6. lawful redescription, fail-closed standing, and the report boundary.

The envelope excludes undeclared curvature couplings, torsion, metric-affine nonmetricity, nonlocal operators, spin/internal carriers, open-system updates, and additional source tensors from the minimal core. Those structures remain admissible as declared richer scopes. The envelope does not assert that any source functional or solution exists.

Proposition 23.3 (No endpoint burden is an anchor). *None of the following may be treated as an anchor of $\mathcal{E}_D$: source-functional existence, covariance, on-shell source closure, positivity or order of active source, nonidentity continuation, matching to Einstein or Schrödinger evolution, role occupancy, or orbit uniqueness.*

Proof. Each listed item changes theorem-bearing output rather than merely fixing the comparison class. It is therefore a tensor, continuation, matching, or realization conclusion. Treating it as an anchor would strengthen the antecedent until the frozen endpoint was assumed. FEP-1.0 forbids that move. $\square$

24 The eight forced certificate roles

Definition 24.1 (Complete gravity–quantum certificate). The certificate type is

$$\mathcal{C}_D^{qq} = (C_{\text{anc}}, C_{\text{ten}}, C_{\text{skin}}, C_{\text{GR}}, C_{\text{QM}}, C_{\text{bdry}}, C_{\text{trans}}, C_{\text{match}}).$$

The coordinates carry, respectively:

1. the exact target anchors;
2. the colimit ledger of standing-bearing tensor, operator, response, source, record, and compatibility content;
3. the maximal congruence invisible to every standing-bearing probe;
4. the AER/Einstein projection object with its constructed source extraction;
5. the projective-Hilbert/Schrödinger projection object with its inertial representation;
6. the record quotient and compatibility quotient together with invoked boundary certificates;

7. an independently constructed upstream RPI continuation object and its closed/open exchange discipline;
8. the canonical comparison cone recovering the declared downstream normal forms.

Definition 24.2 (Forced role set). The target-forced role set is fixed before candidate inspection:

$$\mathcal{R}_D^{gg} = \{r_{\text{anc}}, r_{\text{ten}}, r_{\text{skin}}, r_{\text{GR}}, r_{\text{QM}}, r_{\text{bdry}}, r_{\text{trans}}, r_{\text{match}}\}.$$

The unique-orbit conclusion, realization class, persistence class, and weak-equivalence result are not members of $\mathcal{R}_D^{gg}$.

Theorem 24.3 (No self-reference in the forced role set). *No conclusion depending on the assignment space, live residue, lawful automorphism group, realization licensure, or test-body branch may occupy a role in $\mathcal{R}_D^{gg}$.*

Proof. The role set is fixed before residue and assignment spaces are formed. Using a later orbit or realization conclusion as an occupant would create the cycle

$$\text{assignment conclusion} \rightarrow \text{occupant} \rightarrow \text{assignment conclusion}.$$

The role-occupancy theorem requires role-set and candidate completeness before the orbit is evaluated. Hence endpoint conclusions are downstream only. $\square$

25 Independent role interfaces

The candidate fibres are not generated by attaching eight labels to arbitrary objects. Each fibre is defined by an independently stated universal-property interface.

Definition 25.1 (Role interfaces). For each $r \in \mathcal{R}_D^{gg}$ let lfc_r be the following comparison interface:

1. lfc_{anc} : exact preservation of the anchor tuple $\mathcal{E}_D$;
2. lfc_{ten} : a cocone receiving every declared tensor/source/record/compatibility generator and imposing exactly the declared overlap identifications;
3. lfc_{skin} : the kernel congruence of the joint standing-bearing probe functor;
4. lfc_{GR} : the fixed AER constraint object, Einstein normal form, and a source map satisfying the source-construction interface proved in Part IV;
5. lfc_{QM} : projective state standing, Wigner–Stone continuation, fixed self-adjoint domain, and the scalar Schrödinger normal form;
6. lfc_{bdry} : the product of the measurement-record quotient and the Bell/compatibility quotient, with declared boundary certificates;
7. $\text{lfc}_{\text{trans}}$: an independently formed RPI history/continuation interface whose metric, quantum, record, and compatibility images are proved only afterward;

8. $\text{lfc}_{\text{match}}$: the projection/intertwining interface recovering the independently defined downstream metric, quantum, record, and compatibility continuations from that upstream object.

Definition 25.2 (Role-specific candidate fibres). $\mathcal{O}_r$ is the category of same-target objects satisfying lfc_r , with morphisms the lawful role-preserving isomorphisms of that interface. The broad candidate class is the tagged disjoint union

$$\mathcal{O}_D^{\text{broad}} = \bigsqcup_{r \in \mathcal{R}_D^{gq}} \{r\} \times \text{Ob}(\mathcal{O}_r).$$

The tag records the pre-existing interface domain; it does not establish that the fibre is inhabited or unique.

Definition 25.3 (Presentation equivalence and admissible residue). Within a fibre, $o \sim_{\text{pres}} o'$ iff there is a lawful interface isomorphism that preserves every anchor, standing-bearing probe, continuation, boundary record, and report entry. Cross-fibre identifications are forbidden. The admissible presentation residue is

$$\text{Residue}(\mathcal{O}_D^{\text{pre}}) = \mathcal{O}_D^{\text{adm}} / \sim_{\text{pres}},$$

where admissibility is evaluated against T_D^{gq} and $\mathcal{E}_D$ without assuming the endpoint constructions.

Definition 25.4 (Role signatures). The signatures used to test interface membership are:

$$\begin{aligned} \text{Sig}_{\text{anc}} &= (\text{Target}, \text{Scope}, \text{Reg}, \text{Boundary}, \text{Domains}, \text{Scales}), \\ \text{Sig}_{\text{ten}} &= (\text{Generators}, \text{Overlap}, \text{Cocone}, \text{NoDoubleCount}), \\ \text{Sig}_{\text{skin}} &= (\text{ProbeKernel}, \text{AnchorNeutral}, \text{TensorNeutral}, \text{ContNeutral}), \\ \text{Sig}_{\text{GR}} &= (\text{AER}, \text{SameScopeMetric}_1, \text{SourceConstruction}, \text{EinsteinSolution}), \\ \text{Sig}_{\text{QM}} &= (\text{Ray}, \text{TP}, \text{Stone}, \text{SameScopeQuantumSpatial}_1, \text{Domain}), \\ \text{Sig}_{\text{bdry}} &= (\text{RecordUniversal}, \text{CompatUniversal}, \text{BoundaryCert}), \\ \text{Sig}_{\text{trans}} &= (\text{UpstreamHistory}, \text{Composition}, \text{Nonidentity}, \text{NoRepair}, \text{DirectedExchange}), \\ \text{Sig}_{\text{match}} &= (\text{ProjectionFunctors}, \text{Intertwining}, \text{EinsteinRecovery}, \\ &\quad \text{SchrodingerRecovery}, \text{RecordRecovery}, \text{ReportRecovery}). \end{aligned}$$

The words “construction,” “solution,” and “nonidentity” in these signatures are burdens to be discharged later; they are not target assumptions.

Definition 25.5 (Eligibility relation). For a residue class $x = [(r, o)]_{\sim_{\text{pres}}}$ and a forced role r' , define

$$\text{Elig}_D(x, r') \iff r = r' \text{ and } o \text{ satisfies the independently proved interface } \text{lfc}_{r'}.$$

A positive edge requires a theorem proving the interface. A tag, CSV status, preference, empirical success, or later utility is insufficient.

Proposition 25.6 (Role separation is structural). *A residue eligible for one forced role is not eligible for another unless the certificate grammar itself is amended.*

Proof. The eight interfaces have different universal burdens: fixing the domain, collecting tensor generators, quotienting probe-invisible presentation, realizing a metric projection, realizing a Hilbert

projection, forming boundary quotients, forming an upstream continuation object, and proving its downstream matching/intertwining maps. Cross-eligibility would erase one of those target-fixed type distinctions or permit undeclared work. $\square$

26 Global compatibility before occupation

Definition 26.1 (Global compatibility predicate). For an injective role-covering pre-assignment

$$\tau : \mathcal{R}_D^{gq} \hookrightarrow \text{Residue}_D^{\text{elig}},$$

$\text{Compat}_D(\tau) = 1$ iff:

1. the anchor occupant fixes exactly $\mathcal{E}_D$;
2. all non-anchor occupants refer to the same singleton interior and declared source/clock ledgers;
3. the tensor occupant is the declared colimit and the skin occupant is exactly the joint probe kernel;
4. the GR and QM occupants are downstream and use the response/source constructions proved in Parts IV and VIII;
5. the boundary occupant uses only its universal record and compatibility quotients;
6. the transport occupant is the independently constructed upstream RPI continuation required by Module C;
7. the matching occupant consists of the projection functors and proved intertwining data from that upstream object;
8. the joint ledger is populated without hidden source, update, record, compatibility, or certificate work;
9. all failure and report conditions are preserved.

Lemma 26.2 (Compatibility is not pointwise eligibility). *Pointwise eligibility of eight residue classes does not imply $\text{Compat}_D = 1$.*

Proof. Locally well-typed objects may disagree on interior identity, source extraction, clock, boundary lineage, or solution matching. Those cross-coordinate incidences are not contained in a product of local signatures. $\square$

Definition 26.3 (Acyclic pre-certificate). The pre-certificate is

$$\mathcal{C}_D^{\text{pre}} = (T_D^{gq}, \mathcal{E}_D, \mathcal{R}_D^{gq}, \{\text{Ifc}_r\}_r, \mathcal{O}_D^{\text{broad}}, \sim_{\text{pres}}, \text{Elig}_D, \text{Compat}_D, \text{Failure}_D).$$

It contains no assertion that any difficult interface is inhabited, no live-residue statement, no orbit cardinality, and no realization conclusion.

Theorem 26.4 (Pre-certificate preserves the no-generator theorem). *The pre-certificate is an audit grammar, not a generator of physical occupants.*

Proof. It fixes roles, interfaces, admissibility, equivalence, eligibility, compatibility, and failure types. Every physical or mathematical occupant must still be supplied by an imported theorem-use package or an explicit construction below. $\square$

Theorem 26.5 (No hidden assignment selector). *No score, empirical fit, preferred presentation, or post-hoc relabeling may create an eligibility edge or alter $\mathcal{R}_D^{gg}$.*

Proof. The role set and interfaces are fixed before candidates are inspected. A score changing no interface is skin; one changing an interface adds anchor/tensor data or changes scope. A post-hoc relabeling is target backfitting. $\square$

Part IV

Typed Common Response, Constructed Source, and Composite Closure

27 The full response quotient is not compressed to one number

The mass theorem supplies a quotient-stable deformation-response profile and its transport-facing restriction. The present construction keeps the complete deformation, restoration, composition, scale, and failure data. Galilei mass is extracted only from the transport coordinate; it is not promoted to a complete coordinate of the full response signature.

Definition 27.1 (Full scalar-monopole response presentation). A response presentation is

$$p = (\mathcal{B}, \mu_D^{\text{tot}}(\mathcal{B}), \iota_D^{\text{tr}}(\mathcal{B}), W_D^{\text{full}}, \text{Scale}, \text{Failure}),$$

where $\mathcal{B}$ is an admitted localized or composite persistence lineage, μ_D^{tot} contains every response contribution relevant to that lineage, ι_D^{tr} is its isotropic transport-facing restriction, and W_D^{full} records deformation, restoration, composition, scale-transport, source, and fail-closed witnesses. No Hamiltonian coefficient or metric source is primitive data in p .

Definition 27.2 (Complete response signature). For a presentation p , set

$$\Sigma_D^{\text{full}}(p) = (\text{Deform}, \text{Transport}, \text{Restore}, \text{Compose}, \text{ScaleTransport}, \text{SourceRole}, \text{Failure})_p.$$

Define

$$p \sim_{\text{sc}} p' \iff \Sigma_D^{\text{full}}(p) = \Sigma_D^{\text{full}}(p'), \quad \mathfrak{R}_D^{\text{sc}} = \text{Pres}_D^{\text{full}} / \sim_{\text{sc}}.$$

The quotient is formed upstream of both Hamiltonian and metric-source readouts.

Proposition 27.3 (Quotient stability of full response). $[p]_{\sim_{\text{sc}}}$ is invariant under every lawful change of species name, gauge representative, basis, coordinate, scale coordinate, and witness-neutral source presentation. Independently witnessed restoration or composition differences are not removed merely because they have equal Galilei central charge.

Proof. The first assertion is the inherited MGN1 standing-quotient theorem applied to the full response signature. The second follows because equality of a transport coordinate does not imply equality of the restoration, composition, source-role, or failure coordinates. MGN1 removes unwitnessed proliferation, not independently witnessed distinctions. $\square$

28 Certified composition is indexed by explicit cross-term data

Definition 28.1 (Certified cross-term datum). For a finite family $\vec{x} = (x_i)_{i \in I}$ of response classes, a certified cross-term datum

$$c \in \text{Cross}_D(\vec{x})$$

consists of:

1. an interaction/binding hypergraph on I ;
2. a typed local field or effective carrier for each hyperedge;
3. the corresponding binding, interaction, internal-stress, restoration, transport, and baseline response terms;
4. a no-double-counting incidence map assigning every term to exactly one constituent or hyperedge;
5. lawful-redescription, boundary, scale, and failure certificates for those terms.

Two cross data are equivalent only when their complete cross-response signatures agree.

Definition 28.2 (Typed composite response). For $\vec{x}$ and $c \in \text{Cross}_D(\vec{x})$, define

$$\text{Comp}_D(\vec{x}; c)$$

to be the response class whose signature is the union of the constituent signatures and the certified cross signature, modulo the no-double-counting incidences. The cross datum is an argument. In general there is no binary operation $x \boxplus y$ determined by x and y alone.

Theorem 28.3 (Certified response multicategory). *Response classes, certified cross data, and the operations $\text{Comp}_D(\vec{x}; c)$ form a symmetric multicategory $\mathbf{Resp}_D^{\text{cert}}$:*

1. *the unary empty-cross certificate is the identity;*
2. *permutation transports the hypergraph and leaves the total signature unchanged;*
3. *substitution of certified composites is associative after the induced hyperedge data are flattened and every cross term is retained exactly once.*

No single-valued binary monoid law on $\mathfrak{R}_D^{\text{sc}}$ is asserted.

Proof. The identity certificate adds no cross term. A permutation relabels the finite incidence diagram and changes no role or witness. For substitution, both iterated composites flatten to the same finite decorated hypergraph: constituent vertices, all internal hyperedges, all external hyperedges, and the same no-double-counting map. Equality of that flattened signature gives associativity. The construction remains indexed because different admitted cross certificates for the same constituents may yield different response classes. $\square$

Definition 28.4 (Completed composite residue). Let $\mathfrak{C}_D^{\text{resp}}$ denote the set of equivalence classes of fully specified pairs $(\vec{x}, c)$ under equality of their flattened complete response signatures. Disjoint union of two completed composites is additive only when the cross certificate declares no interaction between the two families; a nontrivial union requires a new cross datum.

29 The inertial transport quotient

Definition 29.1 (Representative inertial charge). For a completed response presentation p , let

$$\hat{m}_i(p) \in \mathbb{R}_{\geq 0}$$

be the central charge of the projective Galilei representation carried by its total isotropic transport coordinate. For a fully specified composite, every binding and interaction contribution to total inertia is included through its cross certificate before the charge is read.

Lemma 29.2 (Transport charge descends but need not separate). *If $p \sim_{\text{sc}} p'$, then $\hat{m}_i(p) = \hat{m}_i(p')$. The converse is not asserted.*

Proof. Equality of the full response signature includes equality of the isotropic transport entry, hence equality of the Bargmann multiplier class and central charge. Presentations with equal central charge may still differ in independently witnessed deformation, restoration, composition, scale, source-role, or failure data. $\square$

Definition 29.3 (Inertial quotient and kinetic map). Let

$$\mathsf{T}_D = \mathfrak{R}_D^{\text{sc}} / \sim_i, \quad x \sim_i y \iff m_i(x) = m_i(y),$$

where $m_i([p]) = \hat{m}_i(p)$. Define the nonneutral inertial domain

$$Q_{\text{in}} := \{x \in \mathfrak{R}_D^{\text{sc}} : m_i(x) > 0\}.$$

Only on this exact domain define

$$q_H : Q_{\text{in}} \longrightarrow \text{Kin}_D, \quad q_H(x) = -\frac{\hbar^2}{2m_i(x)} \Delta.$$

The map factors through the nonneutral part of T_D but does not identify full response classes in its kernel. Neutral response classes remain well typed in $\mathfrak{R}_D^{\text{sc}}$ but do not enter this kinetic formula.

Theorem 29.4 (Constructed inertial representation). *m_i and q_H are well defined on $\mathfrak{R}_D^{\text{sc}}$. For a disjoint certified union,*

$$m_i(\text{Comp}_D(\vec{x}; c_{\text{disj}})) = \sum_i m_i(x_i),$$

and for an interacting composite the same formula holds after the explicit cross contribution $m_i(c)$ is included:

$$m_i(\text{Comp}_D(\vec{x}; c)) = \sum_i m_i(x_i) + m_i(c).$$

Proof. Well-definedness is definition 29.2. Galilei central charges add under independent composition. An interaction or binding term is not omitted: its transport response is an explicit entry of c and contributes its own central charge. The Schrödinger normal-form theorem then supplies the unique kinetic slot in the declared scalar envelope. $\square$

Corollary 29.5 (Transport quotient is one-dimensional, not the full response quotient). *The real span of the image $m_i(\mathfrak{R}_D^{\text{sc}}) \subseteq \mathbb{R}$ is one-dimensional whenever a nonzero inertial class exists. This*

is a theorem only about the inertial transport quotient T_D ; it does not collapse $\mathfrak{R}_D^{\text{sc}}$ or the other response witnesses.

Proof. The image lies in $\mathbb{R}$ and contains a nonzero element. Its real span is therefore one-dimensional. No injectivity of m_i on $\mathfrak{R}_D^{\text{sc}}$ is used. $\square$

Corollary 29.6 (No primitive Hamiltonian mass). *The parameter in $-\hbar^2\Delta/(2m)$ is the transport readout of a fully certified upstream response class. Equal Hamiltonian mass does not authorize identification of response classes that differ in another admitted witness.*

30 Accepted Module A source admission and representation

Accepted theorem object. The source-representation and composite-closure burden in this part is no longer discharged by the rejected finite source grammar of v1.2.0. It is imported from the independently accepted module `MASRC1.0`, manuscript SHA-256 b12ca804f79b881afebbf2ce492b11ac20293d37d2ecb4de454427cc6301cf54; the bounded independent re-review reports every Module A coordinate correct [1, 2]. Only the exact theorem-use interface below is imported.

Definition 30.1 (Capstone identification of the response base). The upstream response quotient used in this capstone is identified with the Module A response base,

$$Q_{\text{resp}} \equiv \mathfrak{R}_D^{\text{sc}}.$$

This identification carries the complete response-protocol signature, admitted deformation acts, incidence data, outputs, support/continuation envelope, lawful redescription, and failure status. It carries no metric source tensor or freely chosen source functional.

Definition 30.2 (Attachment-safe source subtype and structured quotient). For $p \in Q_{\text{resp}}$, Module A first forms the metric-incidence relation from the complete upstream protocol ledger and the tangent of the already licensed faithful metric projection. A complete attachment certificate $\mathfrak{a}_p$ proves that this relation is total, coherent on overlaps, additive, homogeneous, LF-continuous, naturally transported, and metric-protocol complete. A continuous linear functional $\mathcal{W}_p$ attaches only when

$$\text{Attach}_g(p, \mathfrak{a}_p, \mathcal{W}_p) \iff \text{Graph}(\mathcal{W}_p) = \text{MetResp}_p.$$

The genuine source-bearing base subtype and structured source quotient are

$$Q_{\text{resp}}^{\text{src}} = \{[p]_{\text{resp}} : \exists \mathfrak{a}_p, \mathfrak{w}_p \text{ SrcBear}_g(p, \mathfrak{a}_p, \mathfrak{w}_p)\} \subseteq Q_{\text{resp}},$$

$$\widehat{Q}_{\text{src}} = \{(p, \mathfrak{a}_p, \mathfrak{w}_p) : \text{SrcBear}_g(p, \mathfrak{a}_p, \mathfrak{w}_p)\} / \sim_{\text{src}}.$$

There is a surjective forgetful map

$$F : \widehat{Q}_{\text{src}} \twoheadrightarrow Q_{\text{resp}}^{\text{src}}.$$

Zero response is admitted only when the complete independently formed metric-response relation is the zero graph. An upstream class with no complete metric-incidence protocol family has no source

attachment.

Definition 30.3 (Operational source object and action-presentable subclass). The primary metric-facing source object is the complete attached witness

$$\mathcal{W}_x : \Gamma_c(S^2 T^* M) \longrightarrow \mathbb{R}, \quad x \in \hat{Q}_{\text{src}}.$$

Where an action or local source functional $\mathcal{S}_x[g, \Psi]$ independently exists with a separately certified continuous linear first metric variation, it is an action-presentable representative of the same operational witness. Action notation is a compatible subclass and is not required for source admission.

Theorem 30.4 (Accepted attachment-safe metric-source representation). *For every $x \in \hat{Q}_{\text{src}}$, there exists a unique symmetric tensor-density distribution*

$$\mathbf{T}_x \in \mathcal{D}'(S^2 TM \otimes |\Lambda|)$$

such that

$$\mathcal{W}_x(h) = \frac{1}{2} \langle \mathbf{T}_x, h \rangle \quad \text{for all } h \in \Gamma_c(S^2 T^* M).$$

Support, distributional order, lawful-diffeomorphism naturality, regularity class, Ward/exchange status, and declared boundary data are inherited exactly. No stronger regularity is inferred without a regularity certificate.

Proof. This is the accepted Operational Metric-Source Representation Theorem of Module A. The continuous dual of the LF space of compactly supported smooth symmetric covariant tensor fields is the space of symmetric contravariant tensor-density distributions. The attachment theorem supplies the functional from the upstream response protocols before the duality theorem is applied; therefore the distribution represents an independently attached response rather than creating source standing. $\square$

Definition 30.5 (Metric-source tensor class). For $x \in \hat{Q}_{\text{src}}$, write

$$q_g(x) := [\mathbf{T}_x]_{\simeq_{\text{law}}} \in \text{Src}_g^{\text{dist}} / \simeq_{\text{law}}.$$

If a regularity certificate gives $\mathbf{T}_x = T_x dV_g$, the ordinary tensor notation $T_{\mu\nu}[x]$ is used. Boundary-supported components remain explicit coordinates of the source object.

Theorem 30.6 (Accepted quotient descent and probe faithfulness). *The map*

$$q_g : \hat{Q}_{\text{src}} \longrightarrow \text{Src}_g^{\text{dist}} / \simeq_{\text{law}}$$

is well defined and natural. In a common metric envelope,

$$q_g(x) = q_g(y) \iff \mathcal{W}_x(h) = \mathcal{W}_y(h) \quad \text{for every compact metric probe } h,$$

modulo the fixed lawful-redescription and boundary equivalence.

Proof. This is the accepted Module A quotient-descent and probe-faithfulness theorem. Source equivalence transports the complete attachment relation and witness. Equality of all test-section

pairings is equality of the representing tensor distribution; naturality transports that equality across lawful metric presentations. $\square$

Definition 30.7 (Ward and exchange coordinate). Each structured source class retains a distributional current J_x satisfying the complete operational Ward profile

$$\mathcal{W}_x(\mathcal{L}_X g) = -\langle J_x, X \rangle \quad \text{for every compactly supported vector field } X.$$

Closed source status is $J_x = 0$. Nonzero J_x is admissible only as a declared open, boundary, anomalous, or effective exchange coordinate.

Theorem 30.8 (Accepted Ward–divergence equivalence). *For every $x \in \hat{Q}_{\text{src}}$,*

$$\mathcal{W}_x(\mathcal{L}_X g) = -\langle J_x, X \rangle \iff \nabla_\mu \mathbf{T}_{x\nu}^\mu = J_{x,\nu}$$

in the distributional sense. Hence closed source continuation is exactly the case

$$\nabla_\mu \mathbf{T}_{x\nu}^\mu = 0.$$

Proof. This is the accepted Module A Ward-Closure Theorem. Distributional integration by parts gives

$$\frac{1}{2} \langle \mathbf{T}_x, \mathcal{L}_X g \rangle = -\langle \nabla_\mu \mathbf{T}_{x\nu}^\mu, X^\nu \rangle,$$

and the representation identity converts the left side to $\mathcal{W}_x(\mathcal{L}_X g)$. $\square$

Theorem 30.9 (Accepted finite completed-composite source closure). *Let $\mathcal{B}$ be a finite completed source certificate whose source terms are all attachment-certified, whose registry covers each declared term exactly once, whose sector–channel incidence coefficients B_{ae} are explicit, whose external currents are declared, and whose boundary and regularity certificates are fixed. If*

$$\nabla_\mu \mathbf{T}_{a\nu}^\mu = J_{a,\nu}^{\text{ext}} + \sum_e B_{ae} j_{e,\nu}$$

for each sector and every internal incidence column is balanced,

$$\sum_a B_{ae} = 0,$$

then the total source distribution

$$\mathbf{T}_{\mathcal{B}} = \sum_a \mathbf{T}_a$$

satisfies

$$\nabla_\mu \mathbf{T}_{\mathcal{B}\nu}^\mu = \sum_a J_{a,\nu}^{\text{ext}}.$$

In closed scope the right side is zero. An unbalanced internal channel leaves a nonzero residual and cannot be reported as closed.

Proof. This is the accepted Module A Composite-Closure Theorem. Summing the sector Ward equations gives a coefficient $\sum_a B_{ae}$ for each internal current j_e ; column balance cancels it exactly.

Exact term coverage and nonduplication are independent certificate requirements and are verified separately from Ward status. $\square$

Proposition 30.10 (Inhabited accepted source classes). *The accepted module contains both non-source-bearing response presentations and attachment-certified source presentations, including a certified zero response, a smooth Klein–Gordon source, a distributional worldline source, and finite internal-exchange composites. These examples establish nonemptiness of the admitted source interfaces; they do not assert global PDE existence for every source class.*

Remark 30.11 (Module A tensor source and Module B scalar bridge). Module A constructs the complete metric-facing tensor distribution and its Ward/exchange status. It does not collapse that tensor into a scalar mass coordinate. Accepted Module B adds a scalar comparison only on the operator-presentable stable scalar matching subtype, using the attached total-EMT zero-transfer form factor and a controlled Wigner-to-Bargmann contraction. Tensor anisotropy, spin, multipoles, finite-size form factors, and state-dependent expectation energy remain distinct.

31 Exact common-response domain after accepted Modules A and B

Definition 31.1 (Mass-bearing overlap). The common base domain remains

$$Q_{\text{mass}} := Q_{\text{in}} \cap Q_{\text{resp}}^{\text{src}}.$$

Because the metric-source map acts on the structured source refinement, define

$$\hat{Q}_{\text{mass}} := F^{-1}(Q_{\text{mass}}) \subseteq \hat{Q}_{\text{src}}.$$

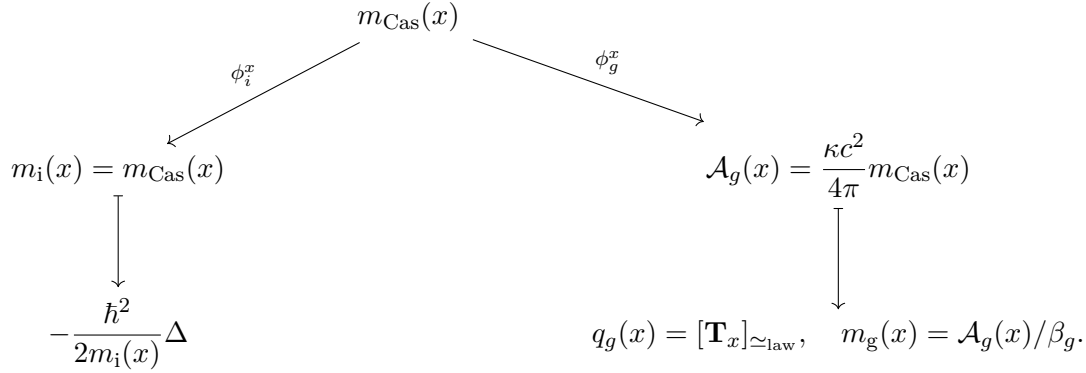
Accepted Module B then restricts to the operator-presentable stable scalar refinement

$$\hat{Q}_{\text{mass}}^{\text{op}} \longrightarrow \hat{Q}_{\text{mass}}$$

and to the exact matching subcategory $\text{Match}_D^{\text{mass}}$, whose objects carry the complete M1–M10 Poincaré–Noether/form-factor/Breit/Bargmann certificate. No claim is made that every mass-bearing source belongs to this subtype.

Theorem 31.2 (Accepted common deformation-response descent). *For every object $x \in \text{Match}_D^{\text{mass}}$, the Hamiltonian inertial and intrinsic active metric-source scalar branches are faithful downstream*

representations of one source-attached stable-sector Noether/Casimir charge $m_{\text{Cas}}(x)$. More precisely,



The upper charge is obtained from the operator realization of the same accepted Module A source attachment. The left branch is the explicit Wigner–Bargmann contraction; the right branch is the exact total-EMT zero-transfer normalization followed by the Einstein first-response monopole. The Hamiltonian operator and tensor source retain distinct codomains.

Proof. Accepted Module A supplies the attachment-safe structured source and tensor distribution. Accepted Module B constructs a normalized-ray family of concrete Module A sources, its homogeneous vector extension, and the polarization-reconstructed operator total EMT. The attached operator charge equals the Stone translation generator on the full reference representation; the stable scalar sector supplies m_{Cas} . The exact Ward normalization $A(0) = 1$ fixes the active zero-transfer branch, while the common-core Wigner contraction fixes the inertial branch. These are precisely the conclusions authorized by definition 15.1. $\square$

Corollary 31.3 (Universal projection-wide scalar comparison). *Within $\text{Match}_D^{\text{mass}}$,*

$$m_g(x) = \alpha_D m_i(x), \quad \alpha_D = \frac{\kappa c^2}{4\pi\beta_g} > 0,$$

with the same α_D for every admitted stable scalar species and completed stable scalar composite. Under the common Einstein–Newton/Bargmann calibration, $m_g = m_i = m_{\text{Cas}}$.

32 No independent same-interior scalar discriminator in the matching envelope

Definition 32.1 (Same-interior mass discriminator). A same-interior mass discriminator is an asserted object-dependent scalar distinction between the accepted active and inertial readouts while preserving the upstream response class, source attachment, attached operator EMT, stable Casimir sector, zero-transfer normalization, contraction, Einstein coupling, boundary data, and common calibration.

Theorem 32.2 (No unwitnessed same-interior mass discriminator). *Within the fixed Module B matching envelope, any additional scalar distinction is one of:*

1. a change in the accepted source attachment or attached operator EMT;

2. *a new total-EMT form factor, source/coupling, boundary, contraction, or calibration witness;*
3. *declared spin, multipole, finite-size, instability, radiation, open-system, or other richer-scope load;*
4. *lawful presentation or normalization skin;*
5. *an unsupported selector or standing failure.*

No independent sixth same-domain scalar discriminator survives.

Proof. The accepted common-charge theorem fixes both scalar readouts from the same attached total-EMT Casimir line, with $A(0) = 1$, one Einstein coupling, one canonical contraction, and one common calibration. A changed ratio must therefore alter a load-bearing witness or leave the envelope. If no witness changes, the distinction has no standing-bearing effect. $\square$

Corollary 32.3 (No witness-free species conversion factor). *A family $m_g^{(a)} = \alpha_a m_i^{(a)}$ in the same matching envelope satisfies $\alpha_a = \alpha_D$ for every admitted object unless an independently declared load-bearing witness changes. This is a theorem of the accepted Module B envelope, not a universal claim about all matter sectors.*

Part V

Accepted Module C: Upstream Continuation and Projection Intertwining

Accepted status. Module C has passed its bounded independent Cycle-3 review and is authorized for capstone import. The two reviewer-recommended editorial clarifications have been implemented without changing any theorem, hypothesis, conclusion, dependency, or endpoint: the full-ray carrier's standard smooth projective-Hilbert-manifold structure is stated explicitly inside the regular local RPI chart, and cube composition is described as whiskered coordinatewise pasting. The accepted Module C object closes G5 in its declared regular local closed/declared-open scope. Accepted Module D now supplies the dependent fibre, live-residue, and unique-orbit closure without reopening Module C.

33 Upstream formation before projection

Definition 33.1 (Raw upstream history). Fix one certified local physical-clock representative τ on a regular local chart of the lawful RPI quotient. A raw upstream history is a clock-forward Moore segment of the smooth RPI-level seed supplied by Module C, carrying the complete upstream standing, localization, field/persistence, deformation-response, transport, record, causal, failure, and declared-exchange signature. No Einstein solution, Hilbert flow, Hamiltonian, or downstream exchange cell occurs in this definition.

Definition 33.2 (Accepted upstream continuation category). The accepted continuation category is the fixed-clock Moore category

$$\mathbf{Cont}_D^\tau,$$

with objects the standing-bearing states in the regular local RPI chart and morphisms the lawful equivalence classes of clock-forward coherent Moore segments. Composition is endpoint-compatible concatenation and identities are zero-duration constant segments.

Theorem 33.3 (Fixed-clock continuation category). $\mathbf{Cont}_D^\tau$ is a category. Coherent composition preserves standing, a failed prefix cannot be repaired by a later segment of the same lineage, and every nonzero-duration segment in the complete packets is nonidentity.

Theorem 33.4 (Nonvacuous upstream formation). *Module C constructs an explicit globally smooth seed directly on a complete local RPI packet and exhibits positive-duration nonidentity morphisms before any downstream projection is defined.*

Proof. These are the accepted fixed-clock Moore-category and complete-packet conclusions of definition 16.1. The seed is RPI-level and clock normalized. In the closed packet it advances both the clock and retained record while leaving the full-ray coordinate fixed upstream; downstream quantum evolution is introduced only by the clock-conditioned projection. $\square$

Proposition 33.5 (No downstream-pullback substitution). *The accepted upstream continuation is formed from the fixed-clock RPI packet and seed before the metric, quantum, record, and compatibility processes are evaluated. A pullback or synchronization of already solved downstream histories may serve as a comparison object only; it cannot replace $\mathbf{Cont}_D^\tau$ or authorize its standing.*

Theorem 33.6 (Clock-presentation equivalence). *If $\tau' = f \circ \tau$ is an orientation-preserving lawful clock change with the corresponding rescaled seed, Module C constructs a duration-transport equivalence*

$$\mathfrak{T}_f : \mathbf{Cont}_D^\tau \longrightarrow \mathbf{Cont}_D^{\tau'}.$$

A nonlinear clock change is therefore a functor between two separately formed presentation categories, not an automorphism of one fixed Moore category.

34 Independently complete role fibres

Definition 34.1 (Metric comparison category). $\mathbf{Met}_D$ is the lawful metric/source orbit category whose objects carry the metric, compact test tensors, accepted Module A source data, gauge and boundary data, Ward/exchange status, Cauchy system, and initial data, transported together under lawful metric redescription. A local gauge section is used only to express the reduced Einstein–Module-A Cauchy system.

Definition 34.2 (Quantum comparison category). $\mathbf{Hilb}_D$ is the fixed projective-Hilbert category determined by the complete full-ray transition profile, exact process, generator graph and domain, spatial/source ledger, baseline, and closure status. Within the regular local RPI chart the abstract full-ray carrier is equipped with its standard smooth projective-Hilbert-manifold structure; this is chart regularity only and carries no upstream authority.

Theorem 34.3 (Independent fibre completeness). *The metric/source, quantum, record, and compatibility fibres used by Module C are independently complete in their fixed targets. Equality of complete profiles is equivalent to lawful role equivalence; only after those completeness theorems may undetected differences be classified as skin.*

35 Descended processes and projection functors

Theorem 35.1 (Projection functors and descended processes). *After $\mathbf{Cont}_D^\tau$ has been formed, Module C defines the closed projection functors*

$$\Pi_g^0, \quad \Pi_H^0, \quad \Pi_{\text{rec}}^0, \quad \Pi_{\text{comp}}^0$$

and their descended processes

$$\widehat{\beta}_g^0, \quad \widehat{\beta}_H^0, \quad \widehat{\beta}_{\text{rec}}^0, \quad \widehat{\beta}_{\text{comp}}^0.$$

Regular metric and quantum profiles descend by smooth projectability. Record and compatibility profiles descend by exact preservation of their final quotient fibres. Identities, composition, standing, and declared failure status are preserved.

Theorem 35.2 (Metric Cauchy intertwining). *On the smooth Cauchy-eligible subtype, the metric projection of an already formed upstream history is lawfully equivalent to the independently defined Einstein–Module-A Cauchy development:*

$$\Pi_g^0 \circ f \simeq_{\text{law}} \widehat{\beta}_g^0(f) \circ \Pi_g^0.$$

The proof uses a lawful metric/source orbit, a local gauge section only for the reduced equations, the seed compatibility equation, and local Cauchy uniqueness after source system, gauge, regularity, boundary, and initial data have been fixed.

Theorem 35.3 (Quantum full-ray intertwining). *On the saturated quantum-eligible subtype,*

$$\Pi_H^0 \circ f \simeq_{\text{law}} \widehat{\beta}_H^0(f) \circ \Pi_H^0.$$

The descended process is a strongly continuous, transition-preserving full-ray group. Wigner and Stone classify its representative implementation, and the accepted spatial normal-form theorem supplies the declared Schrödinger Hamiltonian. The unbounded Stone generator is downstream and is never used as the upstream smooth seed.

Theorem 35.4 (Record intertwining). *The record projection preserves the final measurement-record quotient, fixation, lineage, downstream reuse, and report status under every accepted upstream history.*

Theorem 35.5 (Compatibility intertwining). *The compatibility projection preserves the final joint-standing quotient and its no-transport/no-source-transfer status. No compatible history retypes boundary compatibility as transport, branch selection, or repair.*

36 Distinct closed prediction and actual exchange continuation

Definition 36.1 (Directed exchange cell). For each role $X \in \{g, H, \text{rec}, \text{comp}\}$, Module C distinguishes the counterfactual closed pair

$$(\Pi_X^0, \widehat{\beta}_X^0)$$

from the actual exchange-extended pair

$$(\widetilde{\Pi}_X, \widetilde{\beta}_X)$$

on the enriched carrier $\widetilde{Y}_X = Y_X \times \mathcal{J}_X$. One common upstream incidence $J_D(f) \in \mathcal{J}_D$ restricts to the role incidence $\rho_X J_D(f)$ and defines a directed, generally noninvertible exchange cell

$$\mathcal{E}_X(f) : z_X^0(f) \longrightarrow z_X^1(f).$$

Exchange is tensor/ledger content and is never quotient presentation equivalence.

Theorem 36.2 (Source-target-correct exchange cocycle). *For composable histories f_1, f_2 ,*

$$\mathcal{E}_X(f_2 \circ f_1) = \mathcal{E}_X(f_2) \circ \hat{\beta}_X^0(f_2)(\mathcal{E}_X(f_1)).$$

All role exchanges are restrictions of one common upstream incidence assignment.

37 The accepted continuation four-cube

Definition 37.1 (Continuation four-cube). For one upstream history and every

$$\varepsilon = (\varepsilon_g, \varepsilon_H, \varepsilon_{\text{rec}}, \varepsilon_{\text{comp}}) \in \{0, 1\}^4,$$

define the vertex

$$\mathcal{C}_f(\varepsilon) = \prod_X z_X^{\varepsilon_X}(f)$$

in the product action category. An edge changing only coordinate X is $\mathcal{E}_X(f)$ whiskered by identities in the other factors.

Theorem 37.2 (Cubical coherence and functoriality). *Every two-dimensional face commutes, every pair of monotone paths with the same endpoints has the same composite, and the cube assignment respects history composition. The long-history cube is obtained by whiskered coordinatewise pasting of the short-history cubes. If $J_D(f) = 0$, all role edges are identities and the cube commutes strictly.*

Proof. Exchange edges in distinct role coordinates act in distinct factors of a product action category, so the interchange law gives every square face. Monotone paths differ by adjacent swaps of independent whiskered edges. Additivity of the common incidence and the role cocycles gives the cube of a composite history by whiskered coordinatewise pasting. $\square$

Theorem 37.3 (Complete closed and declared-open inhabitants). *Module C supplies:*

1. *a complete closed nonidentity local RPI packet with a globally smooth upstream seed, lawful vacuum metric/source orbit, clock-conditioned full-ray quantum projection, record and compatibility quotients, and a strict zero-incidence cube;*
2. *a complete declared-open fixed-background packet with one smooth upstream ledger coordinate, a registered common incidence, role restrictions, two composable histories, and verified nonzero exchange cocycles.*

The open packet is not reported as nonzero interacting quantum gravity.

Theorem 37.4 (No deferred certificate repair). *A history that fails standing, source attachment, projection eligibility, record fixation, compatibility typing, or a registered exchange condition at an earlier stage cannot be made the same admissible history by later matching data or a later cube.*

Corollary 37.5 (Gate G5 closed). *The frozen upstream-continuation and projection-commutation gate G5 is closed in the exact accepted Module C envelope. The former solved-development pullback remains only a comparison object and is not used as the foundational upstream continuation.*

Part VI

Complete Joint-Ledger Closure after Accepted Modules A–D

Accepted closure status. Independently accepted Modules A–C supply the metric-source, finite composite-source, operator/source, common-charge, active-readout, inertial-readout, naturality, calibrated-equality, upstream-continuation, projection-intertwining, record, compatibility, directed-exchange, and cubical-coherence coordinates. Independently accepted Module D supplies the source-locked tensor completion, induced joint probe, maximal lawful skin kernel, exact eight-fibre finality, live-residue totality, one lawful assignment orbit, and certificate-relative realization licensure. The ledger is therefore complete for the exact frozen target; no claim is made across changed source, state, seed, potential, or matching targets.

38 Ledger type

Definition 38.1 (Complete joint-ledger grammar). The frozen ledger type is

$$\mathcal{L}_D^{gq} = (L_g, L_H, L_{\text{cross}}, L_{\text{rec}}, L_{\text{comp}}, C_{\text{cert}}, L_{\text{cont}}, L_{\text{report}}).$$

The coordinates respectively record metric source, quantum continuation, common-response incidence, record fixation, compatibility, invoked certificates, upstream continuation with directed exchange, and exact report status. A coordinate is inhabited only by an accepted theorem object or a later independently accepted module.

39 Accepted metric-source coordinate

Definition 39.1 (Metric-source ledger). For an attachment-certified class $\hat{x} \in \widehat{Q}_{\text{src}}$, Module A supplies

$$L_g^A(\hat{x}) = (\hat{x}, \mathbf{a}_{\hat{x}}, \mathcal{W}_{\hat{x}}, \mathbf{T}_{\hat{x}}, J_{\hat{x}}, q_g(\hat{x}), \text{Supp}, \text{Reg}, \text{Boundary}).$$

For a finite completed composite certificate it additionally supplies the exact term registry, signed internal-exchange incidence matrix, declared external currents, total tensor distribution, and closed/open report status.

Theorem 39.2 (Accepted metric-source ledger theorem). *The metric-source coordinate is inhabited on $\widehat{Q}_{\text{src}}$. Its tensor distribution is unique for the attached operational witness, natural under lawful*

metric redescription, and satisfies

$$\nabla_\mu \mathbf{T}^\mu{}_\nu = J_\nu$$

in the distributional sense. For a balanced finite completed composite,

$$\nabla_\mu \mathbf{T}^\mu{}_{B\nu} = \sum_a J_{a,\nu}^{\text{ext}}.$$

Closed scope is the special case in which the declared external sum vanishes.

40 Accepted quantum and common-response coordinates

Definition 40.1 (Quantum ledger). For an admitted inertial class $x \in Q_{\text{in}}$, the quantum ledger is

$$L_H(x) = (\mathcal{H}, \mathbb{P}(\mathcal{H}), \text{TP}, D(H_x), m_i(x), q_H(x), \Pi_H^0, \hat{\beta}_H^0, \text{Mode}_H, \text{Boundary}_H).$$

For an object in the accepted Module B matching subtype,

$$m_i(x) = m_{\text{Cas}}(x), \quad H_{\text{kin}} = -\frac{\hbar^2}{2m_{\text{Cas}}(x)}\Delta.$$

Accepted Module C supplies the clock-conditioned full-ray process and exact Wigner–Stone–Schrödinger intertwining from one upstream history.

Definition 40.2 (Accepted common-response ledger). On $\text{Match}_D^{\text{mass}}$, the accepted common-response coordinate is

$$L_{\text{cross}}^{AB}(x) = (\hat{x}, \hat{\mathbf{T}}_x, \hat{Q}_x^\mu, m_{\text{Cas}}(x), q_H(F\hat{x}), q_g(\hat{x}), m_g(x), m_i(x), \alpha_D).$$

It is a span from one source-attached upstream mass-bearing object and one attached stable-sector charge into distinct metric and Hamiltonian codomains. It is not a GR-to-QM map.

Theorem 40.3 (Accepted common-response ledger theorem). *For every $x \in \text{Match}_D^{\text{mass}}$,*

$$m_i(x) = m_{\text{Cas}}(x), \quad m_g(x) = \alpha_D m_i(x), \quad \alpha_D = \frac{\kappa c^2}{4\pi\beta_g} > 0.$$

The same factor applies to every admitted stable scalar species and completed stable scalar composite in the fixed envelope. Common calibration gives $m_g = m_i = m_{\text{Cas}}$. This theorem does not assert a universal relation outside $\text{Match}_D^{\text{mass}}$.

41 Accepted record and compatibility coordinates

Definition 41.1 (Record ledger).

$$L_{\text{rec}} = (\text{Rec}, \text{ARef}/\sim_{\text{MR}}, \text{FixComm}, \text{Lineage}, \text{Reuse}, \text{Cert}_{\text{rec}}, \Pi_{\text{rec}}^0, \hat{\beta}_{\text{rec}}^0, \text{Report}_{\text{rec}}).$$

Theorem 41.2 (Accepted record-continuation theorem). *Every standing-bearing record use factors through the measurement-record quotient, and Module C proves that the quotient, fixation, lineage,*

reuse, and report status descend from the same accepted upstream history.

Definition 41.3 (Compatibility ledger).

$$L_{\text{comp}} = (\text{BellWitness}, \text{JointStand}, \text{TensorCompat}, \text{LocalRealization}, \\ \text{NoTransport}, \text{Cert}_{\text{comp}}, \Pi_{\text{comp}}^0, \widehat{\beta}_{\text{comp}}^0).$$

Theorem 41.4 (Accepted compatibility-continuation theorem). *Bell-scope standing is carried by joint compatibility rather than hidden transport, source transfer, branch selection, or repair. Module C proves exact compatibility-quotient descent from the same upstream history.*

42 Accepted continuation and exchange coordinate

Definition 42.1 (Continuation ledger). The continuation coordinate is

$$L_{\text{cont}}^C = (\text{Cont}_D^T, \text{ClockTransport}, \text{NoRepair}, \Pi_X^0, \widehat{\beta}_X^0, \widetilde{\Pi}_X, \widetilde{\beta}_X, \mathcal{J}_D, \rho_X, \mathcal{E}_X, \mathcal{C}_f).$$

It records the independently formed fixed-clock RPI history category, lawful clock-presentation equivalences, the four closed projection processes, the four exchange-extended actual processes, one common incidence monoid, directed role restrictions, the exchange cocycles, and the sixteen-vertex continuation cube.

Theorem 42.2 (Accepted continuation-ledger theorem). *The continuation coordinate is inhabited in the exact Module C envelope. It contains nonidentity upstream morphisms, strict closed intertwining, declared-open directed exchange, source-target-correct cocycles, and coherent whiskered coordinatewise pasting of composable cubes.*

43 Certificate and report coordinates

Definition 43.1 (Auxiliary certificate family). C_{cert} contains only certificates with explicit data, hypotheses, use loci, and failure consequences. In this release it includes:

1. Module A attachment, support, naturality, Ward, regularity, boundary, term-registry, and exchange-balance certificates;
2. Module B normalized-ray source lift, homogeneous extension, polarization attachment, charge-generation, stable-sector, form-factor, Breit, Laue, contraction, naturality, and calibration certificates;
3. Module C clock, seed, flow, Moore-category, fibre-completeness, metric-Cauchy, full-ray quantum, record, compatibility, incidence, exchange-cocycle, cubical-coherence, and inhabitant certificates;
4. Module D source-lock, primary-interface, lawful-completion, tensor-colimit, induced-probe, skin-kernel, fibre-finality, live-residue, automorphism, and role-occupancy certificates;
5. the imported operator-domain and test-body certificates actually invoked.

Proposition 43.2 (No empty certificate rhetoric). *A certificate name without data and a proof role does not inhabit C_{cert} .*

Definition 43.3 (Report ledger). L_{report} assigns to each claim the intersection of the scopes of its accepted premises and constructions. It distinguishes accepted, certificate-relative, conditional, declared-open, and nonclaim status.

Theorem 43.4 (Current report-ledger closure). *The present release reports accepted Module A source admission, representation, Ward closure, and finite composite closure; accepted Module B operator/source attachment, common charge, active and inertial scalar readouts, naturality, universal factor, and calibrated equality in the fixed stable scalar envelope; accepted Module C upstream continuation, metric/quantum/record/compatibility intertwining, directed exchange, and strict/declared-open four-cube coherence; accepted Module D tensor/skin construction, eight-fibre exhaustion, live-residue closure, unique lawful assignment orbit, and certificate-relative realization licensure; and the conditional TB1–TB11 weak-equivalence implication.*

44 Complete joint-ledger closure

Theorem 44.1 (Complete joint-ledger inhabitation). *Accepted Modules A–D inhabit every coordinate of the joint-ledger grammar in their exact scopes. The accepted Module D assignment is a compatible bijection from the eight forced roles onto the independently formed live residue, and the complete ledger has one lawful assignment orbit. Therefore the metric, quantum, cross-response, record, compatibility, certificate, continuation, and report coordinates form one complete certificate-relative joint ledger for the frozen target.*

Proof. Modules A–C supply the coordinate objects and their dynamical incidences. Module D supplies target-relative candidate, quotient, eligibility, compatibility, failure, fibre, live-residue, and orbit completeness. Apply definitions 47.4 and 70.3. $\square$

Part VII

Accepted Module D: Complete Certificate Inhabitation

Modular status. Modules A–C supply the accepted source, mass-charge, continuation, projection, record, compatibility, exchange, and matching theorem objects. Accepted Module D constructs the completed tensor and skin targets, proves exact target-relative fibre finality, forms the live residue from all eligible compatible injections, and proves one lawful complete assignment orbit. This part integrates those accepted conclusions; it does not infer them from role tags or from the number of displayed coordinates.

45 Canonical target diagram and eight accepted occupants

Definition 45.1 (Modules A–D canonical occupant tuple). The exact frozen target diagram contains the following eight accepted target-relative classes:

$$\begin{aligned}
C_{\text{anc}}^{ABCD} &= (T_D^{gq}, \mathcal{E}_D, K_D, A_D^+, I_D, \text{ClockScope}, \text{Domains}, \text{Scales}, \text{Failure}), \\
C_{\text{ten}}^D &= L_D^0 \cong \text{colim } \mathbb{G}_D, \\
C_{\text{skin}}^D &= S_D^0 = \text{KerPair}(\text{Probe}_{\Theta_D^{ABC}}), \\
C_{\text{GR}}^{ABCD} &= (\text{AERGravity}_D, \widehat{Q}_{\text{src}}, q_g, \widehat{\mathbf{T}}, A(0) = 1, \Pi_g^0, \widehat{\beta}_g^0, G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}), \\
C_{\text{QM}}^{ABCD} &= (\mathbb{P}(\mathcal{H}), \text{TP}, D(H), m_{\text{Cas}}, m_i, q_H, \Pi_H^0, \widehat{\beta}_H^0, H = -\hbar^2 \Delta / (2m_{\text{Cas}}) + M_V), \\
C_{\text{bdry}}^{ABCD} &= (\Pi_{\text{rec}}^0, \widehat{\beta}_{\text{rec}}^0, \Pi_{\text{comp}}^0, \widehat{\beta}_{\text{comp}}^0, \text{ARef} / \sim_{\text{MR}}, \text{JointStand} / \sim_{\text{comp}}), \\
C_{\text{trans}}^{ABCD} &= (\mathbf{Cont}_D^\tau, \text{ClockTransport}, \text{NoRepair}, \widetilde{\Pi}_X, \widetilde{\beta}_X, \mathcal{E}_X, \mathcal{C}_f), \\
C_{\text{match}}^{ABCD} &= (\phi_g, \phi_i, \text{EinsteinIntertwining}, \text{StoneIntertwining}, \text{RecordRecovery}, \text{CompatibilityRecovery}).
\end{aligned}$$

The tensor and skin coordinates are not imported as assumptions: Module D constructs them from the source-locked accepted Modules A–C interfaces before candidate inspection.

Definition 45.2 (Canonical complete assignment). Let

$$\sigma_0 : \mathcal{R}_D^{gq} \longrightarrow \text{Residue}_D^{\text{live}}$$

map the eight forced roles to the eight accepted lawful residue classes represented in definition 45.1. Accepted Module D proves that this map is a pointwise eligible, globally compatible bijection onto the independently formed live residue.

46 Coordinate inhabitation and target-relative finality

Lemma 46.1 (Anchor fibre). *The anchor role is inhabited by the exact unloaded target tuple. Any different target, scope, regularity, boundary, domain, scale, clock, or failure anchor defines a different target instance rather than a same-target rival.*

Lemma 46.2 (Tensor/operator/source fibre). *The tensor role is inhabited by the explicitly constructed colimit ledger L_D^0 . Every element is generated by accepted Modules A–C primary data through lawful incidence paths; identifications are exactly those generated by the declared overlap relations; and every lawful cocone admits one unique mediator.*

Lemma 46.3 (Skin fibre). *The skin role is inhabited by S_D^0 , the lawful kernel pair of the complete joint probe. It is the unique maximal same-target congruence invisible to every standing-bearing anchor, tensor, projection, continuation, record, compatibility, failure, and report probe.*

Proposition 46.4 (Skin cannot repair accepted Modules A–D data). *No lawful presentation change may alter source attachment, tensor-colimit incidence, operator/source ancestry, the attached total-EMT charge, zero-transfer normalization, the Wigner contraction, the fixed-clock history class, metric/source orbit, full-ray process, record lineage, compatibility class, exchange cocycle, cubical coherence, live-residue membership, or report status.*

Proof. Every listed coordinate is detected by the complete joint probe or fixes the exact target parameter. A change is therefore target motion, tensor/source/continuation motion, failure, or richer scope, not skin. $\square$

Lemma 46.5 (GR/source fibre). *The fixed AER/Einstein/source object is the unique lawful presentation class after the source system, gauge, regularity, initial data, boundary data, time interval, and exchange status are fixed.*

Lemma 46.6 (Quantum fibre). *The fixed projective process, carrier, self-adjoint domain, potential/source ledger, initial ray, boundary data, and clock-conditioned flow determine one lawful Wigner–Stone–Schrödinger presentation class.*

Lemma 46.7 (Boundary fibre). *The fixed record-reuse profile and compatibility-witness profile determine the final measurement-record and compatibility quotient classes used by the certificate.*

Lemma 46.8 (Transport fibre). *The fixed clock, RPI seed, failure boundary, incidence package, and exchange data determine one lawful Module C Moore-history and continuation-cube presentation class.*

Lemma 46.9 (Matching fibre). *The fixed accepted source/charge, metric, quantum, record, compatibility, exchange, and continuation endpoints determine one lawful presentation class of the accepted matching cone.*

47 Eligibility, compatibility, and complete inhabitation

Proposition 47.1 (Coordinate eligibility). *Each class $\sigma_0(r)$ satisfies the independently fixed interface burden for its role and is faithful to the exact frozen target point. Generic interface objects with different source, potential, state, initial data, seed, or matching data remain legitimate objects of different target fibres; they are not collapsed into the frozen target.*

Lemma 47.2 (No anchor conflict). *All eight occupants descend from the same unloaded target and the exact accepted Modules A–D source lock. No source, charge, continuation, colimit, skin, residue, orbit, or licensure conclusion is inserted as an upstream anchor.*

Lemma 47.3 (Global incidence compatibility). *The complete twenty-five-edge target incidence diagram commutes for the canonical assignment. Source attachment, tensor completion, projection maps, common charge, metric and quantum intertwiners, record and compatibility descent, directed exchange, failure status, and report status are jointly consistent.*

Theorem 47.4 (Complete Certificate Inhabitation). *The canonical assignment is a total compatible bijection onto the independently formed live residue:*

$$\sigma_0 : \mathcal{R}_D^{qq} \xrightarrow{\sim} \text{Residue}_D^{\text{live}}, \quad \forall r \text{ Elig}(\sigma_0(r), r), \quad \text{Compat}(\sigma_0) = 1.$$

Every forced role is occupied, every live target-relative residue is used, and no additional same-target live residue remains.

Proof. This is the accepted Module D canonical-assignment, fibre-finality, incidence-compatibility, and live-residue package [7, 8]. Tensor and skin inhabitation follow only after their explicit constructions; the remaining six fibres use their fixed-data accepted uniqueness theorems. The live residue is formed from all eligible compatible injections before its equality with the canonical image is proved. $\square$

Part VIII

Accepted Module B Integration: Common Mass Charge and Inertial–Active Proportionality

Import status. Module B, MBMASS_{1.0}, was judged mathematically correct in bounded independent Cycle 4 under its unchanged standalone endpoint. The sole remaining production defect was a TikZ-CD rendering regression. The exact authorized four-command patch has been applied, the corrected page visually inspected, and the corrected PDF embedded in this project. No theorem, definition, equation, proof, scope condition, or dependency relation was changed by that patch.

48 Exact accepted matching envelope

Definition 48.1 (Accepted Module B matching envelope). The accepted matching category $\text{Match}_D^{\text{mass}}$ consists of objects in the operator-presentable source-attached mass subtype $\hat{Q}_{\text{mass}}^{\text{op}}$ carrying the exact M1–M10 certificate:

1. an accepted Module A mass-bearing source attachment and common metric/source anchors;
2. a normalized-ray family of concrete Module A source presentations formed before lawful quotienting;
3. the norm-squared homogeneous vector extension and its quadratic/continuity certificate;
4. the unique polarization-reconstructed total operator EMT $\hat{\mathbf{T}}_x$ on a common dense invariant domain;
5. a full irreducible Poincaré reference representation whose attached EMT charge implements translations;
6. an isolated stable positive-mass spin-zero spectral sector;
7. regular total-EMT form factors with exact zero-transfer normalization;
8. a regulated stationary Breit transition source and linearized Einstein first-response envelope;
9. an explicit common-core Wigner-to-Bargmann contraction;

10. fixed projection-wide metric and inertial calibrations with lawful transport data.

The envelope does not contain the universal factor, calibrated equality, passive role, or any later capstone conclusion as an assumption.

Proposition 48.2 (No totalization outside the envelope). *Module B does not claim that every element of $\hat{Q}_{\text{mass}}$ is operator-presentable, stable, spin-zero, Poincaré covariant, Breit-regular, or contraction-admissible. Its endpoint is exact on $\text{Match}_D^{\text{mass}}$ and nowhere else.*

49 Operator/source ancestry from accepted Module A

Theorem 49.1 (Normalized-ray source lift and operator attachment). *For every $x \in \text{Match}_D^{\text{mass}}$, there exists a normalized-ray family of concrete accepted Module A source presentations on one fixed spacetime, boundary, regularity, normalization, and improvement representative. Its norm-squared homogeneous extension satisfies the quadratic and joint-continuity certificate and therefore reconstructs, by complex polarization, a unique Hermitian operator-valued total energy–momentum distribution*

$$\hat{\mathbf{T}}_x : \Gamma_c(S^2 T^* M) \longrightarrow \mathcal{L}(\mathfrak{D}_x, \mathfrak{D}'_x).$$

For every normalized $\psi \in \mathfrak{D}_x$,

$$\langle \psi, \hat{\mathbf{T}}_x(h)\psi \rangle = \langle \mathbf{T}_{x, [\psi]}^{\text{ray}}, h \rangle.$$

The operator inherits the accepted Module A support, covariance, Ward, boundary, improvement, and integrated-charge ledgers. An unrelated operator current cannot satisfy the attachment relation.

Proof. This is the operator/source-attachment theorem imported through definition 15.1. Its type separation is essential: the physical source family is defined on normalized rays, while only the auxiliary norm-squared vector extension is polarized. $\square$

Corollary 49.2 (Attached source and operator charges coincide). *Whenever the integrated charge is defined,*

$$\langle \psi, \hat{Q}_x^\mu \psi \rangle = P_{\text{src}, x_\psi}^\mu$$

for every vector in the common core, with off-diagonal charge forms fixed by the same polarization identity.

50 Global charge and stable mass Casimir

Definition 50.1 (Attached total-EMT translation charge). Let $\hat{Q}_x^\mu$ be the global charge of the attached $\hat{\mathbf{T}}_x$ in the full reference representation. The M3–M4 certificate requires its unitary group and the Stone translation representation to implement the same translation automorphisms on the full local observable algebra, with one invariant baseline vector.

Theorem 50.2 (Global source-charge/generator equality). *Under the M3–M4 certificate,*

$$\hat{Q}_x^\mu = \hat{P}_x^\mu$$

on the full reference representation. The proof is global: the two unitary groups differ only by a commutant phase, irreducibility reduces that difference to a scalar, and the invariant baseline removes it. Only afterward is the equality restricted to the stable scalar sector.

Definition 50.3 (Stable Casimir mass). On the isolated positive-mass spin-zero spectral sector,

$$-\eta_{\mu\nu}\hat{P}_x^\mu\hat{P}_x^\nu = m_{\text{Cas}}(x)^2c^2I.$$

This operator Casimir is not identified with $-\eta_{\mu\nu}\omega(\hat{P}^\mu)\omega(\hat{P}^\nu)$ for a generic state, nor with a generic localized-profile energy.

51 Active zero-transfer metric-source branch

Definition 51.1 (Total-EMT form-factor readout). For covariantly normalized generalized spin-zero states in the attached stable sector, write

$$\langle p'|\hat{\mathbf{T}}_x^{\mu\nu}(0)|p\rangle = 2\bar{P}^\mu\bar{P}^\nu A_x(t) + \frac{1}{2}(\Delta^\mu\Delta^\nu - \eta^{\mu\nu}\Delta^2)D_x(t).$$

The integrated attached translation charge fixes the exact zero-transfer Ward normalization

$$A_x(0) = 1.$$

Theorem 51.2 (Attached active metric monopole). *The regulated Breit transition source formed from the same $\hat{\mathbf{T}}_x$ is stationary and distributionally conserved, satisfies*

$$\int d^3r \mathcal{T}_{R,x}^{00} = m_{\text{Cas}}(x)c^2, \quad \int d^3r \mathcal{T}_{R,x}^{ij} = 0,$$

and produces the linearized Einstein first-response monopole

$$h_{00}(r) = \frac{\kappa c^2 m_{\text{Cas}}(x)}{4\pi r} + O(r^{-2}).$$

The D-term, lawful improvements, regulator profile, binding data, and internal-stress distribution do not introduce an independent $1/r$ scalar factor. The active source coordinate is

$$m_g(x) = \frac{\kappa c^2}{4\pi\beta_g} m_{\text{Cas}}(x).$$

Proof. Exact total-EMT charge normalization gives $A_x(0) = 1$. The Breit transition source converts the zero-transfer matrix element into a stationary conserved distribution without requiring one normal state to be simultaneously sharp in momentum and spatially localized. Module A ensures that all certified constituent, binding, mediator, interaction, internal-stress, restoration, transport, baseline, and boundary contributions belong to the same total tensor. $\square$

52 Inertial Bargmann–Schrödinger branch

Theorem 52.1 (Controlled Wigner-to-Bargmann contraction). *For the massive scalar Wigner representation in the Module B certificate, unitary transport to a fixed $L^2(d^3p)$ carrier and restriction to the common Schwartz core yield graph and commutator convergence*

$$E_c(\mathbf{p}) - m_{\text{Cas}}c^2 \longrightarrow \frac{|\mathbf{p}|^2}{2m_{\text{Cas}}},$$

$$K_i^{(c)} \longrightarrow i\hbar m_{\text{Cas}} \partial_{p_i}, \quad [G_i, P_j] = i\hbar \delta_{ij} m_{\text{Cas}} I.$$

Hence the Bargmann central mass and the Schrödinger inertial coefficient are

$$M_B = m_{\text{Cas}} I, \quad H_{\text{kin}} = -\frac{\hbar^2}{2m_{\text{Cas}}} \Delta, \quad m_i(x) = m_{\text{Cas}}(x).$$

53 Common charge, naturality, and one factor

Theorem 53.1 (Accepted common intrinsic charge line). *For every $x \in \text{Match}_D^{\text{mass}}$, the active and inertial scalar branches factor through one attached stable-sector Noether/Casimir charge line:*

$$\begin{array}{ccc} & m_{\text{Cas}}(x) & \\ \swarrow \phi_i^x & & \searrow \phi_g^x \\ m_i(x) = m_{\text{Cas}}(x) & & \mathcal{A}_g(x) = \frac{\kappa c^2}{4\pi} m_{\text{Cas}}(x). \end{array}$$

The full tensor source and Hamiltonian operator remain different theorem objects. Only their intrinsic scalar zero-transfer/Casimir readouts are compared.

Theorem 53.2 (Source and readout naturality). *Lawful morphisms in $\text{Match}_D^{\text{mass}}$ preserve the upstream response identity, normalized-ray Module A source family, coherent concrete representatives, operator attachment, total-EMT charge, stable Casimir sector, zero-transfer normalization, canonical contraction, and projection-wide calibrations. The common charge, inertial, and active source lines therefore form positive-line functors, and the readout maps are natural isomorphisms.*

Corollary 53.3 (One source-conversion anchor). *Within the fixed matching envelope, the Einstein coupling κ , Green/Fourier normalization, and source calibration β_g carry no object index. No independent species-indexed source-conversion factor survives unless a load-bearing source, form-factor, boundary, contraction, or calibration witness changes.*

Theorem 53.4 (Universal inertial–active proportionality). *For every $x \in \text{Match}_D^{\text{mass}}$,*

$$m_g(x) = \alpha_D m_i(x), \quad \alpha_D := \frac{\kappa c^2}{4\pi\beta_g} > 0.$$

The same α_D applies to every stable scalar species and every completed stable scalar composite in the frozen matching envelope.

Proof. The inertial branch gives $m_i = m_{\text{Cas}}$. The active branch gives $m_g = (\kappa c^2 / 4\pi\beta_g)m_{\text{Cas}}$. All coefficients are fixed projection anchors. The ratio is therefore one projection-wide scalar. $\square$

Corollary 53.5 (Active-source order is derived). *Because $\alpha_D > 0$, the positive order of the inertial charge line transports to the active source line. Source order is a consequence of the accepted comparison, not an input to it.*

Corollary 53.6 (Common calibrated equality). *Under the common Einstein–Newton/Bargmann calibration*

$$\beta_g = \frac{\kappa c^2}{4\pi},$$

one has

$$m_g(x) = m_i(x) = m_{\text{Cas}}(x)$$

throughout the calibration-connected fixed envelope.

54 Transport extension and exact boundary

Theorem 54.1 (Flat calibration-transport extension). *If every connected component of the declared extension envelope meets $\text{Match}_D^{\text{mass}}$ and relative calibration transport has trivial holonomy, the natural comparison extends uniquely from the exact matching category to that component. Transport preserves an already proved relation; it does not create matching where the source, stable-sector, or contraction certificate is absent.*

Module B does not prove a generic global energy in curved spacetime, strong equivalence, self-force, all-order self-gravity, spin or multipole equality, unstable or radiating sectors, or a numerical mass spectrum. The Breit object is a transition-source distribution, not a classical probability density or generic normal-state expectation source.

55 Passive role after accepted Module C

Definition 55.1 (Passive role target). The passive role is the participation of the same source-attached localized lineage in free metric continuation with no nongravitational exchange. Module B proves the active–inertial common charge and proportionality. Accepted Module C now constructs the upstream RPI continuation and proves the metric, quantum, record, and compatibility intertwiners. The passive free-body conclusion remains separately conditional on the declared TB1–TB11 test-body certificate.

Definition 55.2 (No independent passive scalar target). Accepted Module C supplies the free-continuation and metric-intertwining carrier. Accepted Module D proves, at the final target-relative certificate-fibre level, that an additional passive scalar either changes the accepted source/coupling/continuation witness, introduces declared nonminimal tensor load, changes the target or scope, or has no standing. This conclusion depends on Modules C–D and does not follow from Module B alone.

Definition 55.3 (Three-role closure target). The frozen final endpoint combines the accepted active–inertial factorization from Module B, the passive/free-continuation intertwining from Module C, and the accepted target-relative fibre/occupancy closure from Module D. Numerical equality is stated only after common calibration, and free-fall consequences only under TB1–TB11.

56 Violation taxonomy after Module B

Theorem 56.1 (Bounded violation classification). *Within the accepted active–inertial envelope, an apparent violation is at least one of:*

1. *a changed accepted Module A source attachment or operator/source incidence;*
2. *a changed total-EMT form factor, Einstein coupling, boundary, contraction, or calibration witness;*
3. *omitted binding, field, interaction, stress, restoration, transport, baseline, or boundary content;*
4. *declared spin, finite-size, multipole, instability, radiation, external exchange, nonminimal coupling, or other richer-scope load;*
5. *failure of the stable-sector, contraction, continuation, or test-body certificate;*
6. *presentation skin, unsupported selector, or standing failure.*

An observed deviation therefore attacks an explicit premise; it is not excluded by terminology.

Part IX

Declared Test-Body Scope and Weak-Equivalence Closure

57 Why the motion theorem is separate

Common-role descent and universal source proportionality do not by themselves prove that localized bodies follow geodesics. Geodesic motion is a theorem in a metric projection under additional analytic and energy hypotheses. The capstone therefore isolates the conventional bridge and states every premise visibly. None is imported under the generic words *standing*, *body*, or *free fall*.

58 The conventional Geroch–Jang theorem

The bridge chosen for the core weak-equivalence result is the Geroch–Jang theorem [29, 30]. Ehlers–Geroch, Gralla–Wald, and Geroch–Weatherall provide stronger or complementary small-body formulations and are used as comparators and extension routes [31, 32, 33].

Theorem 58.1 (Geroch–Jang motion theorem, imported). *Let (M, g_{ab}) be an orientable relativistic spacetime and let $\gamma : I \rightarrow M$ be a smooth embedded curve. Suppose that for every open set O containing $\gamma[I]$, there exists a smooth symmetric tensor field T^{ab} such that:*

1. T^{ab} satisfies the strengthened dominant energy condition;
2. $\nabla_a T^{ab} = 0$;
3. $\text{supp}(T^{ab}) \subset O$;
4. T^{ab} is nonzero at some point of O .

Then γ is timelike and can be reparametrized as a geodesic.

Remark 58.2 (Energy condition is load-bearing). The strengthened dominant energy condition is not decorative. Conservation, localization, and nontriviality without an appropriate energy condition do not force timelike geodesic motion. The capstone therefore treats the energy condition as an explicit test-body anchor rather than a consequence of the kernel.

59 AASC test-body certificate

Definition 59.1 (Declared test-body certificate). For a smooth embedded curve γ in the metric projection, $\text{TestBodyCert}_D(\gamma)$ holds iff:

- TB1.** γ lies in the smooth orientable Lorentzian metric envelope A_g ;
- TB2.** for every open neighborhood $O \supset \gamma[I]$, the source architecture admits a smooth symmetric metric-facing tensor T_O^{ab} ;
- TB3.** T_O^{ab} is the extraction of a localized nonzero composite response class $[\mu_D^{\text{tot}}(\mathcal{B}_O)]_{\sim\mu}$;
- TB4.** $\text{supp}(T_O^{ab}) \subset O$;
- TB5.** T_O^{ab} is nonzero somewhere in O ;
- TB6.** $\nabla_a T_O^{ab} = 0$ in the declared free closed scope;
- TB7.** T_O^{ab} satisfies the strengthened dominant energy condition;
- TB8.** no nongravitational force, open-system exchange, hidden boundary flux, or update acts on the body;
- TB9.** the family preserves the same localized body-role lineage as O shrinks;
- TB10.** self-force, spin, multipole, and backreaction terms beyond the leading test-body target are excluded or separately declared;
- TB11.** the report is restricted to the leading timelike test-body conclusion.

Exact conditional status. The capstone does not prove that every physical body satisfies TB1–TB11. It proves that any body/family satisfying this explicit certificate has the stated metric motion. The source and localization roles make the certificate physically typed; the energy and shrinking-support clauses remain additional conventional hypotheses.

60 Hypothesis-by-hypothesis occupation

Proposition 60.1 (Metric and orientation occupation). *TB1 is occupied by the declared Lorentzian metric projection together with orientability as an explicit test-body anchor.*

Proof. The Einstein projection fixes a smooth four-dimensional Lorentzian metric envelope. Orientability is not generated by the kernel and is included only in the test-body certificate because definition 58.1 requires it. $\square$

Proposition 60.2 (Source and symmetry occupation). *TB2–TB3 are occupied when the localized body family admits the complete metric source extraction of definition 30.9 on every neighborhood in the test-body family.*

Proof. Metric variation of the closed source ledger yields a symmetric metric-facing tensor. The source is tied to the total deformation-response class and therefore is not a primitive point-particle scalar. The quantification over neighborhoods is an additional localization-family requirement, not a consequence of one bounded body alone. $\square$

Proposition 60.3 (Support and nontriviality occupation). *TB4–TB5 are exactly the conventional small-support and nonzero-body conditions. They refine RPI bounded localization and non-null persistence but are not identical to those roles.*

Proof. RPI supplies bounded localization and non-null standing-bearing content. The theorem requires the stronger family statement: for every neighborhood of γ , an eligible tensor is supported there and nonzero. The test-body certificate records that strengthening explicitly. $\square$

Proposition 60.4 (Conservation occupation). *TB6 is occupied by the closed metric source ledger when no exchange is declared.*

Proof. By definition 39.2, coherent closed source continuation yields $\nabla_a T_O^{ab} = 0$. If an external force or exchange is present, TB6 is not occupied and the Geroch–Jang theorem is inapplicable; the motion belongs to a forced or open scope. $\square$

Proposition 60.5 (Energy condition status). *TB7 is an independent test-body assumption or sector-specific theorem. It is not derived from admissibility, standing, source conservation, or common deformation-response descent.*

Proof. The kernel governs theorem status, not the sign or causal character of every stress-energy tensor. Conservation does not imply the strengthened dominant energy condition, and known counterexamples show that weaker conditions are insufficient for the full timelike conclusion. Therefore TB7 must be independently certified. $\square$

Proposition 60.6 (Free and leading-order scope). *TB8–TB10 exclude nongravitational exchange and higher-order body effects from the exact target. Their failure does not refute the capstone; it moves the motion problem to a declared forced, self-force, spin, multipole, or backreaction scope.*

61 Geodesic motion

Theorem 61.1 (Certified Test-Body Geodesic Theorem). *If $\text{TestBodyCert}_D(\gamma)$ holds, then γ is timelike and admits a parametrization with tangent u^a satisfying*

$$u^a \nabla_a u^b = 0.$$

Proof. TB1–TB7 reproduce the hypotheses of definition 58.1; TB8–TB11 ensure that the imported conclusion is applied to the intended free leading-order metric scope and is not overreported. Therefore the Geroch–Jang theorem yields a timelike geodesic. $\square$

Remark 61.2 (Why the equation is mass independent). The geodesic equation contains the metric connection and initial tangent, but no scalar inertial or active mass. Mass and internal structure enter the eligibility of the stress-energy family and higher-order corrections, not the leading free trajectory once the certificate is satisfied.

62 Weak-equivalence closure

Definition 62.1 (Weak-equivalence target). The weak-equivalence target in this paper is: any two certified free localized bodies with the same initial event and unit timelike tangent in the same metric projection have the same leading test-body worldline, independently of their calibrated deformation-response scale and species label.

Theorem 62.2 (Weak-Equivalence Closure in the Declared Test-Body Scope). *Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be admitted localized bodies whose shrinking-support families satisfy TestBodyCert_D about curves γ_1 and γ_2 . If the curves have the same initial event and tangent in the same smooth metric projection, then*

$$\gamma_1 = \gamma_2$$

on their common geodesically well-posed interval. The result is independent of m_i , m_g , species label, and internal response scale at leading test-body order.

Proof. By definition 61.1, both curves are timelike geodesics of the same metric. The geodesic initial-value problem has a unique local solution for fixed initial event and tangent. Hence the curves agree on the common interval. No mass parameter appears in the geodesic equation, and TB8–TB11 exclude nongravitational forces, directed exchange, and higher-order body effects from the target. The conditional weak-equivalence conclusion therefore follows independently of the proportionality theorem of Part VIII. That theorem supplies a separate source/inertial representation result, not a premise of Geroch–Jang motion. $\square$

Corollary 62.3 (Passive response closure). *Under TestBodyCert_D , passive free response introduces no new mass role beyond the common deformation-response architecture. It is the metric continuation of the same localized source class.*

63 Relation to stronger small-body results

The Ehlers–Geroch theorem relaxes the fixed-background idealization by treating sufficiently small size and mass in general relativity [31]. Gralla–Wald uses a scaled one-parameter family of metrics and proves that the limiting worldline is a geodesic, with self-force and spin-curvature effects at first perturbative order [32]. Geroch–Weatherall formulates small-body motion using tracking and extends the analysis to field wave packets [33]. These results support richer certificate routes.

Proposition 63.1 (Extension classification). *Replacing the core Geroch–Jang bridge by Ehlers–Geroch, Gralla–Wald, or Geroch–Weatherall requires a new test-body certificate matching the exact hypotheses of the selected theorem. It strengthens the downstream motion analysis but does not alter the kernel, RPI, gravity, common-descent, or certificate-occupancy proofs.*

Proof. The replacement changes the analytic motion bridge and its anchors, not the upstream role architecture. By no-backflow, no such theorem may retro-authorize the interior. Therefore the upstream capstone remains fixed while the conditional metric corollary is enriched. $\square$

64 Nonclaims and boundary of the weak-equivalence result

Nonclaim 64.1 (No strong equivalence principle). The theorem does not prove local equivalence of all nongravitational experiments, universality for arbitrary self-gravitating bodies, or independence from gravitational binding energy at every order.

Nonclaim 64.2 (No all-order self-force). The theorem does not remove gravitational self-force, spin-curvature force, multipole force, radiation reaction, or geodesic-deviation corrections. Those are higher-order or richer-body ledgers.

Nonclaim 64.3 (No nonminimal-coupling exclusion). A declared scalar, vector, torsion, curvature, or species-dependent coupling may alter motion. It changes the source or connection scope and must be evaluated as a richer theorem object.

Nonclaim 64.4 (No empirical guarantee of TB7). The paper does not assert that all physical matter satisfies the strengthened dominant energy condition. The condition is explicit because it is load-bearing for the chosen theorem.

Theorem 64.5 (Test-Body Claim-Status Theorem). *The weak-equivalence result of definition 62.2 is a rigorous conditional theorem within the exact certificate of definition 59.1. It is neither an unconditional theorem for every physical body nor a merely heuristic consequence.*

Proof. Every premise is listed and mapped to an occupant or an explicit additional assumption. The imported mathematical theorem is precise. The conclusion follows deductively. The report ledger prevents omission of the conditions or promotion beyond the declared scope. $\square$

Part X

Accepted Module D: Exact Fibre Exhaustion and Role-Occupancy Closure

Accepted status. Module D has an independent positive Cycle-5 hostile-referee verdict under its unchanged standalone endpoint. This part imports only its target-relative completion theorem. Generic GR, quantum, source, state, potential, seed, and matching moduli remain nontrivial; uniqueness is asserted only for presentations of the exact frozen Modules A–C target instance.

65 Completed antecedent certificate

Definition 65.1 (Completed antecedent certificate). The antecedent certificate C_D^{ABC} contains the exact accepted Modules A–C target data, the source-locked primary tensor generators and overlaps, their probe maps, the frozen report/failure boundary, and no already completed tensor ledger, skin relation, live residue, assignment orbit, or realization conclusion.

Definition 65.2 (Target-fibred comparison categories). For each forced role r , let $\mathcal{M}_r$ be its generic interface moduli groupoid and let $\mathcal{F}_r(\theta_r)$ be the target presentation groupoid over the exact frozen target parameter θ_r . Generic interface eligibility is independent of the intended occupant. Target fidelity is the separate requirement that the candidate lie over θ_r .

66 Candidate, quotient, compatibility, and failure completeness

Theorem 66.1 (Candidate completeness). *Every target-relevant response generated by the fixed certificate is represented by the eight-sorted response grammar: coordinate extraction, lawful redescription, interface restriction, licensed quotient or factorization, incidence evaluation, failure/report formation, or explicit reset to a new target certificate.*

Theorem 66.2 (Presentation quotient and eligibility completeness). *Within each target fibre, lawful interface isomorphism removes exactly presentation multiplicity. The independent role burden and target-fidelity predicates classify every generated residue as eligible for one role, a lawful duplicate, a different target instance, hidden load, or failure.*

Theorem 66.3 (Compatibility and failure completeness). *The complete twenty-five-edge incidence registry contains every generated cross-coordinate co-instantiation condition for the frozen target. Every failed candidate has an explicit earliest failure status; no unresolved selector or untyped remainder remains.*

67 Live residue and assignment space

Definition 67.1 (Compatible injections and live residue). Let Σ_D^{pre} be the set of pointwise eligible injections from the forced role set into the admissible presentation residue that satisfy every compatibility edge. Define

$$\text{Residue}_D^{\text{live}} = \bigcup_{\sigma \in \Sigma_D^{\text{pre}}} \text{im } \sigma.$$

The live residue is formed from all compatible injections before canonical identification.

Proposition 67.2 (Assignment nonemptiness). *The canonical accepted Modules A–D assignment belongs to Σ_D^{pre} .*

68 Eight target-relative fibre theorems

Proposition 68.1 (Anchor fibre). *The anchor target fibre is nonempty and connected; a different anchor tuple is a different target instance.*

Proposition 68.2 (Tensor fibre). *The tensor target fibre is represented uniquely by the explicit colimit L_D^0 of the lawfully completed source-locked generator/overlap diagram.*

Proposition 68.3 (Skin fibre). *The skin target fibre is represented uniquely by S_D^0 , the maximal lawful kernel pair of the complete joint probe.*

Proposition 68.4 (GR/source fibre). *After source system, gauge, regularity, initial data, boundary data, interval, and exchange status are fixed, the GR/source target fibre has one lawful presentation class.*

Proposition 68.5 (Quantum fibre). *After projective process, carrier, Hamiltonian domain, potential/source ledger, initial ray, boundary, and clock data are fixed, the quantum target fibre has one lawful presentation class.*

Proposition 68.6 (Boundary fibre). *The fixed record-reuse and compatibility-witness profiles determine one final boundary target class.*

Proposition 68.7 (Transport fibre). *For the fixed Module C clock, seed, failure boundary, incidence, and exchange package, the upstream continuation target fibre has one lawful Moore-history/cube presentation class.*

Proposition 68.8 (Matching fibre). *The fixed accepted endpoints and intertwining data determine one lawful matching-cone presentation class.*

69 Live-residue identification and role-realization completeness

Theorem 69.1 (Exact live-residue identification).

$$\text{Residue}_D^{\text{live}} = \{\rho_{\text{anc}}, \rho_{\text{ten}}, \rho_{\text{skin}}, \rho_{\text{GR}}, \rho_{\text{QM}}, \rho_{\text{bdry}}, \rho_{\text{trans}}, \rho_{\text{match}}\}.$$

Every complete compatible injection has the same image and uses every live residue class.

Corollary 69.2 (Role-realization completeness). *The frozen certificate is candidate-, quotient-, role-set-, eligibility-, compatibility-, live-residue-, and failure-complete.*

70 Lawful automorphisms and unique orbit

Definition 70.1 (Lawful certificate automorphism). A lawful automorphism preserves the eight role sorts, frozen target parameters, all twenty-five incidence edges, complete joint-probe values, failure/report statuses, and every source-locked ancestry relation.

Proposition 70.2 (Automorphism action). *The lawful automorphism group acts on the final compatible assignment space. For the frozen target, the pairwise distinct role signatures and target parameters make that action trivial.*

Theorem 70.3 (Unique Complete Assignment Orbit). *Let*

$$\Sigma_D^{gg} = \left\{ \sigma : \mathcal{R}_D^{gg} \xrightarrow{\sim} \text{Residue}_D^{\text{live}} : \forall r \text{ Elig}(\sigma(r), r), \text{Compat}(\sigma) = 1 \right\}.$$

Then

$$\left| \Sigma_D^{gg} / \Gamma_D^{\text{lawful}} \right| = 1.$$

Proof. By the eight fibre theorems and definition 69.1, the final assignment space contains exactly the canonical compatible bijection. The lawful role-incidence automorphism action is trivial. This is the accepted Module D one-orbit theorem [7, 8]. $\square$

Corollary 70.4 (Role-occupancy closure). *The exact frozen certificate satisfies Role-Occupancy Closure.*

Theorem 70.5 (Complete Realization Closure). *The role-realization-complete certificate licenses one complete constraint-native realization class of the exact frozen target, unique up to lawful equivalence:*

$$\text{RealizationLicensed}(C_D^{ABC}, T_D^{gg}).$$

Proof. Apply the sufficiency direction of definition 20.1 using definitions 69.2 and 70.3. The conclusion is certificate-relative; changed source, state, potential, initial data, seed, or matching data define a new target instance. $\square$

Part XI

Adversarial Map for the Modules A–D Closure Release

71 Adversarial-map gate

Definition 71.1 (Adversarial-map gate). Before an objection is reported, it must identify the exact target, theorem-use import, accepted Module A, B, C, or D construction attacked; preserve the kernel dependency order; and trace the chain to the earliest false premise, undefined map, invalid inference, or overreported conclusion.

Proposition 71.2 (No objection by formalism preference). *Metric-first, Hilbert-first, action-first, canonical-quantization, path-integral, or empirical-first preference has no referee standing unless it identifies a failed named link in the frozen kernel-first chain.*

72 Accepted Module A attacks

72.1 “Every upstream response is still declared source-bearing”

No. The accepted source type first forms an independently attached operational metric-response relation. The base subtype is

$$Q_{\text{resp}}^{\text{src}} = \{[p]_{\text{resp}} : \exists \mathfrak{a}_p, \mathfrak{w}_p \text{ SrcBear}_g(p, \mathfrak{a}_p, \mathfrak{w}_p)\} \subseteq Q_{\text{resp}},$$

and the domain of the source map is the structured quotient $\hat{Q}_{\text{src}}$. A response with no complete metric-incidence protocol family receives no source attachment.

72.2 “The arbitrary zero functional still passes”

It does not. Zero attaches only when the complete upstream metric-incidence family returns zero for every compactly supported metric variation. The source attachment is graph equality with the independently formed relation MetResp_p , not a freely chosen functional paired with p .

72.3 “The tensor distribution is merely notation”

For an attached continuous linear operational witness $\mathcal{W}$, Module A proves the unique distributional-section representation

$$\mathcal{W}(h) = \frac{1}{2} \langle \mathbf{T}_{\mathcal{W}}, h \rangle.$$

An objection must therefore attack continuity/linearity, the distributional duality theorem, lawful naturality, or representative transport; it cannot merely observe that the tensor is distributional rather than smooth.

72.4 “Ward closure is assumed”

The exact accepted statement is an equivalence:

$$\mathcal{W}(\mathcal{L}_X g) = -\langle J, X \rangle \iff \nabla_\mu \mathbf{T}^\mu{}_\nu = J_\nu.$$

Closed status is the certified case $J = 0$; declared open status retains the named current. Module A does not infer $J = 0$ for every source.

72.5 “Composite closure hides interaction exchange”

The finite completed composite contains an exact term registry and signed sector–channel incidence. Internal cancellation follows only when each internal incidence column sums to zero. An unbalanced channel remains as residual divergence and cannot be reported as closed.

73 Accepted Module B attacks

73.1 “The Casimir is still obtained from a generic state expectation”

It is not. Module B distinguishes the operator Casimir m_{Cas} , the selected-state mean four-momentum, and localized-profile energy. The common-charge theorem uses the stable-sector operator Casimir and the attached total-EMT zero-transfer normalization. A mean-rest wave packet is included as a separating counterexample.

73.2 “The operator EMT can still be unrelated to the accepted Module A source”

It cannot in the accepted matching category. The physical source family is formed on normalized rays at the Module A prequotient level, with one coherent concrete representative family fixed before quotienting. Its norm-squared vector extension is polarized to reconstruct the unique operator EMT. Both the translation charge and form factors must use that attached operator.

73.3 “Projective-state invariance still conflicts with quadratic homogeneity”

The two roles are now separated. The physical source map is defined on normalized rays and is phase invariant. The auxiliary vector-weight extension scales as $|\lambda|^2$ and is used only for polarization. No object is required to satisfy both normalization invariance and nontrivial quadratic scaling.

73.4 “The active monopole omits binding or stress content”

The form-factor branch uses the complete attached total EMT inherited from Module A. Exact term coverage includes certified constituent, binding, mediator, interaction, internal-stress, restoration, transport, baseline, and boundary contributions. The zero-transfer Ward normalization and Laue identity apply to that total tensor, not to a bare constituent sum.

73.5 “Module B proves mass equality in every matter or spacetime regime”

It does not. The theorem is restricted to the operator-presentable stable positive-mass spin-zero matching envelope with exact Poincaré, form-factor, Breit, Einstein first-response, Wigner-contraction, and calibration certificates. Massless, unstable, open, radiating, spinning, multipolar, generic curved-spacetime, and strong-equivalence claims remain outside the import.

74 Accepted Modules C and D attacks

74.1 “The manuscript constructs an upstream GR/QM continuation”

It does in the accepted Module C scope. Module C constructs a fixed-clock RPI Moore-history category before any downstream process, proves lawful clock-presentation equivalence, complete metric/quantum/record/compatibility fibres, strict and declared-open intertwiners, and the sixteen-vertex continuation cube. The former downstream solved-development pullback remains only a rejected comparator.

74.2 “The unique certificate orbit is inferred from role tags”

It is not. Module D fixes the role set before candidate inspection, separates generic interface eligibility from exact target fidelity, constructs the tensor colimit and joint-probe skin kernel before residue formation, forms the live residue from all compatible injections, proves eight target-relative fibre theorems, and only then computes the lawful assignment quotient.

75 Same-scope rival normal forms

Theorem 75.1 (Current rival classification). *Relative to the accepted Modules A–D results, a purported same-target rival certificate construction must differ by at least one of:*

1. *upstream response or attachment incidence;*
2. *operational metric-response graph;*
3. *continuity/linearity, support, regularity, naturality, or boundary data;*
4. *represented tensor distribution;*
5. *Ward/exchange current;*
6. *composite term registry or exchange incidence;*
7. *lawful source-witness equivalence;*
8. *report status;*
9. *a declared richer source scope.*

If none changes, the rival is a lawful representation of the same accepted source object. Changed source, potential, state, initial-data, seed, or matching packages define new target instances and are not collapsed by this same-target theorem.

Proof. The listed source coordinates are retained by $\hat{Q}_{\text{src}}$ and the completed composite certificate; the accepted Module B matching coordinates additionally retain the concrete ray-source family, operator attachment, stable charge, zero-transfer normalization, contraction, and calibration data. A difference outside them changes no accepted Module A probe and is presentation-only; a difference within them changes tensor, boundary, exchange, or scope content. $\square$

76 Attack-summary ledger

Attack	Current standing	Earliest response or open link
Endpoint is target-loaded	Standing if found	Definitions 23.1 and 23.3.
Every response becomes a source	Closed by accepted Module A	Definitions 14.1 and 30.2.
Zero functional can be attached arbitrarily	Closed by accepted Module A	Attachment graph and certified-zero theorem in embedded Module A.
Source tensor lacks quotient descent	Closed on $\hat{Q}_{\text{src}}$	Definitions 30.4 and 30.6.
Composite divergence is assumed	Closed for finite completed certificates	Definition 30.9.
Universal mass factor is unproved	Closed within accepted Module B envelope	Definitions 15.1 and 53.4.
Upstream continuation is downstream pullback	Closed by accepted Module C	Definitions 16.1, 33.2, 33.5 and 37.1.
Unique orbit is asserted from role tags	Closed by accepted Module D	Definitions 17.1, 66.1, 69.1 and 70.3.
Weak equivalence hides analytic hypotheses	Closed conditionally	Definitions 59.1 and 62.2.
Strong equivalence or all quantum gravity is claimed	False reading	Definition 82.1.

Part XII

Modules A–D Integration Theorems and Final Frozen-Endpoint Closure

77 Accepted Module A integration theorem

Theorem 77.1 (Attachment-Safe Metric-Source Integration Closure). *Let D instantiate the unloaded fixed capstone target of definition 23.1, the kernel K_D , consequence layer A_D^+ , singleton realized physical interior I_D , and the theorem-use packages of Part II. Then:*

1. *source-bearing standing is restricted to the genuine subtype $Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$, formed through the independent attachment theorem;*
2. *the structured source quotient $\hat{Q}_{\text{src}}$ retains attachment, complete witness, support, naturality, Ward/exchange, regularity, and boundary data, with a surjective forgetful map $F : \hat{Q}_{\text{src}} \twoheadrightarrow Q_{\text{resp}}^{\text{src}}$;*
3. *every $x \in \hat{Q}_{\text{src}}$ has a unique natural symmetric tensor-density distribution $\mathbf{T}_x$ representing its complete operational metric response;*
4. *the quotient-faithful metric-source map*

$$q_g : \hat{Q}_{\text{src}} \longrightarrow \text{Src}_g^{\text{dist}} / \simeq_{\text{law}}$$

is well defined and probe-faithful;

5. *closed and open source status are exactly*

$$\nabla_\mu \mathbf{T}_{x\nu}^\mu = 0 \quad \text{or} \quad \nabla_\mu \mathbf{T}_{x\nu}^\mu = J_{x,\nu};$$

6. *every finite attachment-certified composite with exact term coverage, no duplication, balanced internal signed incidence, and declared external currents satisfies*

$$\nabla_\mu \mathbf{T}_{B\nu}^\mu = J_\nu^{\text{ext}};$$

7. *arbitrary zero functionals, unattached tensors, and source factors introduced only by labels acquire no source standing.*

Proof. This is the accepted Module A theorem-use package definition 14.1, integrated through definitions 30.4, 30.6, 30.8 and 30.9. □

78 Accepted Module B integration theorem

Theorem 78.1 (Common Mass-Charge and Universal Inertial–Active Proportionality Closure). *Let $x \in \text{Match}_D^{\text{mass}}$ be an object of the accepted Module B stable scalar matching category over an accepted Module A mass-bearing source. Then:*

1. *the normalized-ray Module A source family, coherent concrete representative lift, norm-squared homogeneous extension, and complex-polarization certificate reconstruct a unique attached operator total EMT $\hat{\mathbf{T}}_x$;*
2. *the global charge of $\hat{\mathbf{T}}_x$ equals the Stone translation generator on the full irreducible reference representation;*
3. *the isolated stable scalar sector supplies one positive Poincaré mass Casimir $m_{\text{Cas}}(x)$;*
4. *the exact attached total-EMT zero-transfer normalization is $A_x(0) = 1$, and the regulated Breit transition source produces*

$$h_{00}(r) = \frac{\kappa c^2 m_{\text{Cas}}(x)}{4\pi r} + O(r^{-2});$$

5. *the explicit Wigner-to-Bargmann contraction yields*

$$M_{\text{B}} = m_{\text{Cas}}(x)I, \quad H_{\text{kin}} = -\frac{\hbar^2}{2m_{\text{Cas}}(x)}\Delta;$$

6. *active and inertial scalar readouts factor through the same attached Noether/Casimir charge while their full tensor and operator codomains remain distinct;*
7. *one projection-wide factor satisfies*

$$m_{\text{g}}(x) = \alpha_D m_{\text{i}}(x), \quad \alpha_D = \frac{\kappa c^2}{4\pi\beta_g} > 0;$$

8. *the same factor applies to every admitted stable scalar species and completed stable scalar composite in the frozen matching envelope;*
9. *common Einstein–Newton/Bargmann calibration gives*

$$m_{\text{g}}(x) = m_{\text{i}}(x) = m_{\text{Cas}}(x).$$

Proof. These are the exact conclusions of the independently accepted Module B theorem-use package definition 15.1, integrated through definitions 31.2, 53.1, 53.2, 53.4 and 53.6. The import is restricted to $\text{Match}_D^{\text{mass}}$ and does not promote the theorem to massless, unstable, spinning, multipolar, radiating, open, or generic curved-spacetime sectors. $\square$

79 Accepted Module C integration theorem

Theorem 79.1 (Kernel-Native Upstream Continuation and Cubical Projection Intertwining Closure). *Within the declared regular local fixed-clock Module C envelope:*

1. a certified local clock and a smooth, locally Lipschitz, clock-normalized RPI-level seed are formed before any metric, Hilbert, Hamiltonian, or solved downstream path;
2. the fixed-clock Moore category $\mathbf{Cont}_D^T$ is nonempty, contains nonidentity morphisms, composes strictly, preserves standing, and is fail-closed under prefixes;
3. lawful orientation-preserving clock changes induce duration-transport equivalences between separately formed presentation categories;
4. the metric/source, quantum, record, and compatibility fibres are independently complete in their fixed targets;
5. closed projection functors and descended processes are formed only after the upstream category;
6. on the exact eligible subtype, the metric image intertwines with the independently fixed Einstein–Module-A Cauchy development;
7. on the exact saturated subtype, the full-ray quantum image is strongly continuous, transition preserving, and Wigner–Stone–Schrödinger classified;
8. record fixation/lineage and compatibility-not-transport descend from the same upstream history;
9. distinct closed-prediction and actual exchange-extended processes are related by directed cells generated by one common upstream incidence and satisfy the source-target-correct cocycle;
10. the four role directions form a sixteen-vertex continuation cube whose faces and monotone paths commute by whiskered coordinatewise pasting; zero incidence gives the strict closed cube;
11. a complete closed nonidentity packet and a complete declared-open fixed-background packet inhabit every mandatory Module C coordinate.

Proof. This is the accepted Module C theorem-use package definition 16.1, integrated through definitions 33.3, 33.4, 35.1 to 35.5, 36.2, 37.2 and 37.3. The two accepted editorial clarifications change no mathematical content. $\square$

80 Accepted Module D integration theorem

Theorem 80.1 (Exact Certificate-Fibre Exhaustion and Unique Role-Occupancy Closure). *Relative to the exact frozen Modules A–C target instance, accepted Module D proves:*

1. the source-locked primary tensor interfaces admit lawful covariant cosieve completions in

$$\mathbf{TenInt}_D = \mathbf{Set}^{S_D} / \mathcal{W}_D;$$

2. the completed generator and overlap diagram has the explicit tensor colimit L_D^0 , with no undeclared generator or extra identification;
3. the accepted probes extend by the same adjunction, induce the tensor probe, and assemble with the remaining accepted probes into one complete joint probe;
4. the lawful kernel pair S_D^0 is the unique maximal same-target standing-neutral skin congruence;

5. *the exact eight target-relative certificate fibres are candidate-, quotient-, eligibility-, compatibility-, and failure-complete;*
6. *the live residue is formed from all eligible compatible injections before canonical identification and contains exactly eight lawful residue classes;*
7. *the canonical assignment is a compatible bijection onto that live residue;*
8. *the certificate-lawful role-incidence automorphism action has one assignment orbit:*

$$\left| \Sigma_D^{gg} / \Gamma_D^{\text{lawful}} \right| = 1;$$

9. *the imported Role-Occupancy Closure theorem therefore yields*

$$\text{RealizationLicensed}(C_D^{ABC}, T_D^{gg}).$$

Proof. This is the accepted Module D theorem-use package definition 17.1, integrated through definitions 47.4, 66.1 to 66.3, 69.1, 70.3 and 70.5. The conclusion is target-relative: changed source, state, potential, initial data, seed, or matching packages define a new target instance. $\square$

Corollary 80.2 (Gate consequence of Modules A–D integration). *G1–G5 and G7 are closed in the exact accepted module envelopes. G6 remains separately correct and conditional under TB1–TB11. Consequently the core FEP-1.0 complete-certificate realization endpoint is closed relative to the frozen target.*

81 Conditional weak-equivalence theorem

Theorem 81.1 (Weak-Equivalence Closure in the Declared Test-Body Scope). *Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be localized body families in the faithful metric projection, each satisfying TB1–TB11 of definition 59.1. If their limiting curves share the same initial event and unit future timelike tangent, then both curves are the same timelike geodesic on the common well-posed interval. The leading trajectory is independent of species label and calibrated response scale.*

Proof. This is definition 62.2. It follows from the exact imported small-body hypotheses and local uniqueness of the geodesic initial-value problem. Modules B and C are compatible with the result but are not substituted for the TB1–TB11 hypotheses. $\square$

82 Final scope and reportability

Theorem 82.1 (Modules A–D Integration Scope and Reportability). *The reportable endpoint of v1.6.0 is exactly:*

1. *preservation of the frozen kernel dependency order;*
2. *attachment-safe source-bearing admission, natural distributional metric-source representation, and exact Ward/divergence classification;*

3. *finite completed-composite closure with exact term coverage, nonduplication, and signed internal-exchange cancellation;*
4. *operator/source attachment from normalized-ray accepted Module A source presentations;*
5. *equality of the attached total-EMT charge with the translation generator and exact stable-sector Casimir extraction;*
6. *exact total-EMT zero-transfer active metric readout and explicit Wigner-to-Bargmann inertial readout;*
7. *common attached charge factorization, one projection-wide inertial-active proportionality factor, and common calibrated equality in the frozen stable scalar matching envelope;*
8. *independently formed fixed-clock RPI continuation, lawful clock-presentation transport, and no-repair history closure;*
9. *metric, quantum, record, and compatibility descent and intertwining from the same upstream history;*
10. *distinct closed-prediction and actual exchange-extended processes, directed exchange cocycles, and strict/declared-open sixteen-vertex continuation cubes;*
11. *lawful covariant completion of the source-locked tensor interfaces, explicit tensor-colimit closure, induced joint-probe completion, and maximal lawful skin-kernel closure;*
12. *exact eight-fibre target-relative finality, live-residue closure, one lawful complete assignment orbit, and certificate-relative realization licensure;*
13. *the separately conditional TB1–TB11 weak-equivalence implication.*

The release does not report uniqueness across changed target parameters, strong equivalence, generic curved-spacetime mass, a nonzero interacting quantum-gravity law, all-order self-force or multipole closure, or all quantum-gravity dynamics.

83 Conclusion

The dependency order remains

$$\text{TargetAdeq}_D \Longrightarrow K_D \Longrightarrow A_D^+ \Longrightarrow I_D,$$

then

$$I_D \Longrightarrow (\text{Gravity}_D, \text{QuantumStand}_D) \Longrightarrow (\Pi_g(I_D), \Pi_H(I_D)).$$

The accepted modular chain is

$$Q_{\text{resp}} \Longrightarrow Q_{\text{resp}}^{\text{src}} \Longrightarrow \hat{Q}_{\text{src}} \xrightarrow{q_g} \text{Src}_g^{\text{dist}} / \simeq_{\text{law}},$$

then, on the accepted stable scalar matching subtype,

$$\hat{Q}_{\text{mass}}^{\text{op}} \Rightarrow \hat{\mathbf{T}}_x \Rightarrow m_{\text{Cas}} \Rightarrow \begin{cases} m_g = \alpha_D m_{\text{Cas}}, \\ m_i = m_{\text{Cas}}, \end{cases}$$

then

$$\begin{aligned} \text{RPI seed} &\Rightarrow \mathbf{Cont}_D^\tau, \\ \mathbf{Cont}_D^\tau &\Rightarrow \text{four faithful role processes,} \\ &\Rightarrow \text{strict or exchange-certified continuation cube.} \end{aligned}$$

and finally

$$\begin{aligned} \text{source-locked primary interfaces} &\Rightarrow \text{lawful covariant completions} \Rightarrow L_D^0, \\ L_D^0 &\Rightarrow S_D^0 \Rightarrow \text{Residue}_D^{\text{live}}, \\ \text{Residue}_D^{\text{live}} &\Rightarrow \left| \Sigma_D^{gq} / \Gamma_D^{\text{lawful}} \right| = 1 \Rightarrow \text{RealizationLicensed}(C_D^{ABC}, T_D^{gq}). \end{aligned}$$

The core frozen endpoint is therefore closed relative to the exact accepted target. The weak-equivalence branch remains a separate conditional theorem under TB1–TB11, and no sequel claim is promoted beyond its certified scope.

A Formal Predicates and Type Lock

This appendix records the exact accepted types after integration of Modules A–D. The metric-source, finite-composite, common-charge, active–inertial proportionality, upstream-continuation, projection-intertwining, cubical-coherence, tensor-completion, skin-kernel, fibre-finality, live-residue, and unique-orbit branches now have accepted theorem objects in their declared envelopes.

A.1 Kernel-first target

$$\text{TargetAdeq}_D(T) \Rightarrow K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{lrr}_D) \Rightarrow A_D^+ \Rightarrow I_D.$$

A faithful downstream projection $\Pi : I_D \rightarrow X$ must preserve anchors, standing-bearing content, lawful quotient identity, continuation status, and reportability. No downstream object authorizes an upstream role.

A.2 Response and inertial types

Let RespPres_D be the admitted response presentations and let $\sim_{\text{resp}}$ be equality of the complete upstream protocol-response signature under lawful transport. Then

$$Q_{\text{resp}} = \text{RespPres}_D / \sim_{\text{resp}}.$$

The inertial domain is the separately typed subset

$$Q_{\text{in}} \subseteq Q_{\text{resp}}$$

for which the transport-facing response and the scalar nonrelativistic projection are admitted. On that domain,

$$q_H : Q_{\text{in}} \rightarrow \mathbf{Kin}_D, \quad q_H(x) = -\frac{\hbar^2}{2m_i(x)}\Delta.$$

No converse claim identifies full response classes from equal m_i .

A.3 Attachment-safe source-bearing type

For a response presentation p , Module A forms the metric-incidence and operational metric-response relations

$$\begin{aligned} \text{MetInc}_p(h, \pi) &\iff \exists \delta [\text{Inc}_p(\pi, \delta) \wedge \tau_{g,p}(\delta) = h], \\ \text{MetResp}_p &= \{(h, a) : \exists \pi [\text{MetInc}_p(h, \pi) \wedge \text{Out}_p(\pi) = a]\}. \end{aligned}$$

An attachment certificate $\mathfrak{a}_p$ proves provenance, totality, overlap coherence, linear first-order compatibility, LF continuity, lawful transport, and metric-protocol completeness. The attachment relation is

$$\text{Attach}_g(p, \mathfrak{a}_p, \mathcal{W}) \iff \text{AttCert}_g(p, \mathfrak{a}_p) \wedge \text{Graph}(\mathcal{W}) = \text{MetResp}_p.$$

A complete source witness additionally carries support, naturality, Ward, regularity, and boundary data. Hence

$$Q_{\text{resp}}^{\text{src}} = \{[p]_{\text{resp}} : \exists \mathfrak{a}_p, \mathfrak{w}_p \text{SrcBear}_g(p, \mathfrak{a}_p, \mathfrak{w}_p)\} \subseteq Q_{\text{resp}},$$

while the structured source-witness quotient is

$$\widehat{Q}_{\text{src}} = \{(p, \mathfrak{a}_p, \mathfrak{w}_p) : \text{SrcBear}_g(p, \mathfrak{a}_p, \mathfrak{w}_p)\} / \sim_{\text{src}}.$$

There is a surjective forgetful map

$$F : \widehat{Q}_{\text{src}} \twoheadrightarrow Q_{\text{resp}}^{\text{src}}.$$

A.4 Accepted source representation

For $\hat{x} \in \widehat{Q}_{\text{src}}$, the attached operational witness is a continuous linear functional on compactly supported symmetric metric variations. Module A constructs the unique tensor-density distribution

$$\mathbf{T}_{\hat{x}} \in \mathcal{D}'(S^2TM \otimes |\Lambda|)$$

with

$$\mathcal{W}_{\hat{x}}(h) = \frac{1}{2} \langle \mathbf{T}_{\hat{x}}, h \rangle.$$

The quotient-faithful source map is

$$q_g : \widehat{Q}_{\text{src}} \rightarrow \text{Src}_g^{\text{dist}} / \simeq_{\text{law}}, \quad q_g(\hat{x}) = [\mathbf{T}_{\hat{x}}]_{\simeq_{\text{law}}}.$$

The Ward predicate is

$$\mathcal{W}_{\hat{x}}(\mathcal{L}_X g) = -\langle J_{\hat{x}}, X \rangle \iff \nabla_\mu \mathbf{T}_{\hat{x}}^\mu = J_{\hat{x}, \nu}.$$

A.5 Finite composite closure

A completed finite composite has sector tensors $\mathbf{T}_a$, declared external currents J_a^{ext} , internal channel currents j_e , and signed incidence coefficients B_{ae} satisfying

$$\nabla_\mu \mathbf{T}_{a\nu}^\mu = J_{a,\nu}^{\text{ext}} + \sum_e B_{ae} j_{e,\nu}.$$

Exact internal balance is

$$\forall e \quad \sum_a B_{ae} = 0,$$

which gives

$$\nabla_\mu \left(\sum_a \mathbf{T}_{a\nu}^\mu \right) = \sum_a J_{a,\nu}^{\text{ext}}.$$

A.6 Exact common-response overlap

The base overlap is

$$Q_{\text{mass}} = Q_{\text{in}} \cap Q_{\text{resp}}^{\text{src}},$$

and its structured refinement is

$$\hat{Q}_{\text{mass}} = F^{-1}(Q_{\text{mass}}) \subseteq \hat{Q}_{\text{src}}.$$

For $\hat{x} \in \hat{Q}_{\text{mass}}$, both

$$q_H(F\hat{x}) \quad \text{and} \quad q_g(\hat{x})$$

are defined. Accepted Module B further restricts to the operator-presentable stable scalar matching subtype

$$\hat{Q}_{\text{mass}}^{\text{op}} \supseteq \text{Ob}(\text{Match}_D^{\text{mass}}),$$

constructs one attached Noether/Casimir charge $m_{\text{Cas}}(x)$, and proves

$$m_{\text{i}}(x) = m_{\text{Cas}}(x), \quad m_{\text{g}}(x) = \alpha_D m_{\text{Cas}}(x), \quad \alpha_D = \frac{\kappa \mathcal{C}^2}{4\pi\beta_g}.$$

Common calibration gives $m_{\text{g}} = m_{\text{i}} = m_{\text{Cas}}$ only in that exact envelope.

A.7 Accepted Module C continuation type

Fix one certified local clock representative τ . Module C constructs the fixed-clock Moore category

$$\mathbf{Cont}_D^\tau,$$

from a smooth RPI-level seed before any downstream solution is evaluated. Orientation-preserving clock changes give duration-transport equivalences between separately formed presentation categories.

The four closed projection pairs are

$$(\Pi_X^0, \hat{\beta}_X^0), \quad X \in \{g, H, \text{rec}, \text{comp}\},$$

with smooth projectability for regular metric/quantum profiles and exact quotient preservation for record/compatibility profiles. The metric/source carrier is a lawful groupoid orbit; the quantum carrier is the fixed full-ray profile with its standard smooth projective-Hilbert-manifold structure inside the regular local RPI chart. That smooth representative structure is chart regularity only and has no upstream authority.

The actual exchange-extended pairs are

$$(\tilde{\Pi}_X, \tilde{\beta}_X)$$

on $\tilde{Y}_X = Y_X \times \mathcal{J}_X$. One common incidence $J_D(f) \in \mathcal{J}_D$ restricts by ρ_X and defines the directed cell $\mathcal{E}_X(f)$. The cocycle is

$$\mathcal{E}_X(f_2 \circ f_1) = \mathcal{E}_X(f_2) \circ \hat{\beta}_X^0(f_2)(\mathcal{E}_X(f_1)).$$

For $\varepsilon \in \{0, 1\}^4$,

$$\mathcal{C}_f(\varepsilon) = \prod_X z_X^{\varepsilon_X}(f)$$

forms the accepted continuation four-cube. Composition is by whiskered coordinatewise pasting in the product action category. Zero incidence gives the strict closed cube.

A.8 Accepted Module D completion type

Let $\mathcal{S}_D, \mathcal{W}_D$ be the source-locked tensor-incidence schema and witness/report object. Module D completes each primary interface by the covariant left-Kan construction in

$$\text{TenInt}_D = \mathbf{Set}^{\mathcal{S}_D} / \mathcal{W}_D,$$

forms the completed tensor ledger

$$L_D^0 \cong \text{colim } \mathbb{G}_D,$$

and induces the tensor and joint probes. The lawful skin object is

$$S_D^0 = \text{KerPair}(\text{Probe}_{\Theta_D^{ABC}}).$$

For the eight forced roles, define the compatible injection space and live residue as in definition 67.1. Accepted Module D proves

$$\text{Residue}_D^{\text{live}} = \{\rho_{\text{anc}}, \rho_{\text{ten}}, \rho_{\text{skin}}, \rho_{\text{GR}}, \rho_{\text{QM}}, \rho_{\text{bdry}}, \rho_{\text{trans}}, \rho_{\text{match}}\},$$

then

$$\left| \Sigma_D^{gg} / \Gamma_D^{\text{lawful}} \right| = 1,$$

and finally

$$\text{RealizationLicensed}(C_D^{ABC}, T_D^{gg}).$$

The orbit and licensure statements are final conclusions, not antecedents or certificate occupants. Changed source, state, potential, initial data, seed, or matching parameters define changed target instances.

B Certificate Coordinate and Final Integration-Status Ledger

ID	Coordinate	Accepted occupant	Closure theorem	v1.6 status
C1	Anchor ledger	Exact unloaded target, kernel, RPI, declared metric, Poincaré, stable-sector, fixed-clock, Cauchy, exchange, contraction, calibration, domain, boundary, failure, and report data	Module D target-fibred anchor finality; changed anchor data define a changed target instance	Closed
C2	Tensor/operator/source ledger	Accepted Modules A–C generators and overlaps completed lawfully; explicit colimit ledger L_D^0	Module D covariant cosieve completion and tensor-colimit universal property	Closed
C3	Skin ledger	$S_D^0 = \text{KerPair}(\text{Probe}_{\Theta_D^{ABC}})$	Module D induced joint-probe completeness and maximal lawful kernel-pair theorem	Closed
C4	GR/source projection	AER gravity, Einstein normal form, Module A source map, Module B active readout, Module C lawful metric/source orbit and Einstein–Module-A Cauchy intertwining	Module D fixed-data GR/source fibre finality	Closed
C5	QM projection	Projective standing, Wigner–Stone type, scalar Schrödinger normal form, Module B Bargmann matching, Module C clock-conditioned full-ray process and exact quantum intertwining	Module D fixed-data quantum-fibre finality	Closed
C6	Boundary projection	Measurement-record and compatibility quotients descended from the same accepted upstream history	Module D fixed reuse/witness-profile boundary-fibre finality	Closed
C7	Transport/continuation	Accepted fixed-clock Moore category, clock-presentation equivalences, no-repair closure, closed and actual processes, common incidence, directed exchange, and strict/open four-cube	Module C G5 theorem plus Module D fixed seed/incidence transport-fibre finality	Closed
C8	Matching ledger	Accepted Module B active–inertial scalar matching plus Module C metric, quantum, record, compatibility, and exchange intertwiners	Module D fixed-endpoint matching-cone finality and global incidence closure	Closed

C Theorem-Use Provenance Ledger

Source family	Imported conclusion	Use in v1.6	Prohibited promotion
Kernel	Target adequacy forces Adm/Stand/Ref/Irr.	Governs every target and proof object.	No metric, Hilbert, field, source, charge, continuation, tensor completion, or numerical generation.
Unus/singleton	One same-domain admissible interior up to lawful equivalence.	Blocks an independent I_g/I_H split.	No claim that all richer domains coincide.
RPI	Coherent, role-closed, operationally minimal physical interior.	Common upstream role architecture and the domain from which Module C histories are formed.	Not a source functional, Hamiltonian, Einstein solution, or assignment orbit.
AER/gravity	Coisotropic standing, characteristic redescription, first-class closure, multiplier form.	Gravitational role type.	Not the Einstein equation or all gravitational dynamics.
Einstein	Minimal closed-source metric normal form.	Downstream metric equation type once source and Cauchy data are supplied.	No numerical κ, Λ ; no all-gravity theorem.
Quantum and Schrödinger	Projective standing, Wigner–Stone continuation under its hypotheses, and $a\Delta + M_V$ normal form.	Quantum projection type and inertial representation on Q_{in} .	No claim that every RPI history has this image without Module C eligibility.
Mass/R9	μ_D and ν_D as deformation and transport response.	Upstream response role.	No gravitational equality or numerical spectrum by itself.
MGN1	Witness/quotient discipline and no free numerical selector.	Blocks unwitnessed promotion of coordinates, normalizations, and species factors.	Does not eliminate independently witnessed source directions or derive a mass value.
Accepted Module A	Attachment-safe source subtype; tensor-distribution representation; lawful descent; Ward/divergence equivalence; finite composite exchange cancellation.	Closes metric-source representation and finite composite-source closure; supplies $Q_{\text{resp}}^{\text{src}}, \hat{Q}_{\text{src}}, q_g, \mathbf{T}$, and J .	No common charge, upstream continuation, or assignment orbit.
Module A positive review	Independent authorization of Module A.	Source theorem import gate.	Does not authorize later modules.

Source family	Imported conclusion	Use in v1.6	Prohibited promotion
Accepted Module B	Normalized-ray source lift; operator/source attachment; attached total-EMT charge; $A(0) = 1$ active readout; explicit Wigner contraction; common charge line; one factor and calibrated equality.	Closes the common-charge branch of G2 and G3 on $\text{Match}_D^{\text{mass}}$.	No massless, unstable, spinning, multipolar, radiating, or generic curved-space extension.
Module B positive Cycle-4 review	Independent mathematical authorization plus a typesetting-only patch authorization.	Mass-charge theorem import gate.	Does not authorize Module C or D.
Accepted Module C	Fixed-clock RPI Moore histories; lawful clock-presentation equivalence; complete four-role fibres; Einstein and Wigner–Stone–Schrödinger intertwiners; record/compatibility descent; directed exchange; sixteen-vertex cube; strict and declared-open inhabitants.	Closes G5 and supplies the transport and matching objects.	No global preferred clock, nonzero interacting quantum gravity, or fibre orbit by itself.
Module C positive Cycle-3 review	Independent authorization plus two non-fatal editorial clarifications.	Continuation theorem import gate.	Does not authorize Module D.
Accepted Module D	Source-locked covariant tensor completion; explicit tensor colimit; induced joint probe; maximal skin kernel; eight target-relative fibre theorems; live-residue closure; one lawful assignment orbit; Role-Occupancy licensure.	Closes G1 and G7 and completes the core FEP-1.0 certificate-relative realization endpoint.	No uniqueness across changed source, state, seed, potential, or matching targets; no new dynamics.
Module D positive Cycle-5 review	Independent authorization of the completed Module D endpoint.	Final certificate import gate for v1.6.	Does not promote sequel claims or remove TB1–TB11.
Records	Record quotient and fixation.	Module C record descent and Module D boundary-fibre finality.	No hidden outcome selector.
Entanglement	Joint standing compatibility and no hidden transport.	Module C compatibility descent and Module D boundary-fibre finality.	No metric communication or source transfer.
Trajectory	Coherence, composition, fail-closed prefixes, and no deferred repair.	Module C fixed-clock history category.	Does not choose a physical seed.
Transport	Standing-preserving transport geometry.	Module C projection and exchange vocabulary.	No primitive metric or connection authority.

Source family	Imported conclusion	Use in v1.6	Prohibited promotion
Role occupancy	One compatible assignment orbit licenses realization under exact completeness hypotheses.	Applied after Module D candidate, quotient, eligibility, compatibility, failure, live-residue, and orbit closure.	No generator from cardinality, role tags, or an incomplete candidate set.
Structural unification	Paired projection, joint-ledger necessity, route exhaustion.	Specification of the certificate coordinates and failure routes.	Not by itself the complete realization.
Geroch–Jang	Exact localized conserved energy-condition family implies a timelike geodesic.	Conditional TB1–TB11 bridge.	No claim every body satisfies its hypotheses.

D Test-Body Hypothesis Map

Conventional hypothesis	AASC/certificate carrier	Status	Failure classification
Smooth orientable Lorentzian spacetime	A_g plus orientability in A_{TB}	Declared anchor	Different metric/test-body scope
Smooth embedded curve	Test-body localization family	Declared target datum	No well-typed limiting curve
Smooth symmetric T^{ab}	Complete metric source extraction	Derived when source/domain regularity holds	Incomplete source or regularity failure
Conservation $\nabla_a T^{ab} = 0$	Closed metric ledger	Derived in closed scope	External force, open exchange, or incomplete ledger
Support in every neighborhood	Shrinking-support test-body family	Additional assumption/certificate	Not a test-body limit
Nonzero somewhere	Non-null localized persistence	Strengthened to each neighborhood	Null body / vacuum source
Strengthened dominant energy condition	TB7	Independent assumption or sector theorem	The timelike geodesic theorem unavailable
No nongravitational force	TB8 and zero exchange ledger	Declared scope condition	Forced motion / open system
Same lineage under shrinking	Reference and trajectory certificate	Declared compatibility condition	Different bodies across the family
Leading-order scope	TB10	Declared report anchor	Higher-order self-force, spin, multipole, or backreaction scope
Geodesic conclusion	Geroch–Jang theorem	Imported theorem	No conclusion if a premise fails

Conventional hypothesis	AASC/certificate carrier	Status	Failure classification
Mass-independent uniqueness	Geodesic IVP plus universal coupling	New corollary	Different initial data or altered connection

D.1 Comparator routes

- **Ehlers–Geroch**: useful when small size and mass are treated with metric backreaction rather than a fixed background source family.
- **Gralla–Wald**: useful for scaled one-parameter families and leading self-force/spin corrections.
- **Geroch–Weatherall**: useful for tracking formulations and field wave packets.

Each replacement requires its own exact certificate. None changes the kernel dependency.

E Adversarial Attack Map

Objection	Status	Successful attack burden	Named link
Kernel is optional	Conditional	Preserve the same determinate target while omitting one kernel role without governance-equivalent replacement	Definitions 2.2 and 4.2
RPI already proves the capstone	Conditional	Exhibit in RPI the accepted A–D source, charge, continuation, tensor/skin, and unique-orbit theorems	Definitions 14.1, 15.1, 16.1 and 17.1
Endpoint is target-loaded	Standing if found	Identify source existence, common charge, continuation, tensor colimit, unique orbit, or WEP as an antecedent anchor	Definitions 23.1 and 23.3
Every response is declared source-bearing	Closed by Module A	Break attachment independence, source-subtype formation, or exact graph equality	Definitions 14.1 and 30.2
Arbitrary zero source passes	Closed by Module A	Exhibit a response with no total metric-incidence family that nevertheless satisfies attachment	Embedded Module A certified-zero theorem
Distributional source is not unique	Conditional	Break the continuous-dual representation or lawful source equivalence	Definitions 30.4 and 30.6
Composite internal exchange is hidden	Closed for finite completed certificates	Exhibit duplicate/unregistered terms or a balanced incidence whose internal divergence does not cancel	Definition 30.9
Universal inertial/source factor is unproved	Closed in the exact Module B envelope	Break operator/source attachment, total-EMT charge, $A(0) = 1$, contraction, naturality, or calibration	Definitions 15.1 and 53.4

Objection	Status	Successful attack burden	Named link
Upstream continuation is a downstream pull-back	Closed by Module C	Show the fixed-clock RPI history category or seed is defined from solved downstream histories	Definitions 16.1, 33.2 and 33.5
Projection descent is merely assumed	Closed within Module C scopes	Break one fibre-completeness, process-descent, Cauchy, Wigner–Stone, record, or compatibility theorem	Definitions 34.3 and 35.2 to 35.5
Declared exchange is hidden inside equivalence	Closed by Module C	Show a nonzero exchange is quotiented as skin or that the cocycle is ill typed	Definitions 36.1, 36.2 and 37.1
Tensor ledger is asserted rather than constructed	Closed by Module D	Break the covariant cosieve completion, completed overlap diagram, or colimit universal property	Definitions 17.1 and 46.2
Skin collapses standing-bearing content	Closed by Module D	Exhibit a joint-probe-visible difference identified by S_D^0 , or break joint-probe completeness	Definitions 46.3 and 46.4
Unique orbit follows from role tags	Closed by Module D	Break candidate, quotient, eligibility, compatibility, failure, fibre, live-residue, or automorphism completeness	Definitions 66.1, 69.1 and 70.3
Weak equivalence hides energy/support hypotheses	Conditional	Show one TB1–TB11 premise is omitted or misapplied	Definitions 59.1 and 62.2
Strong equivalence or all quantum gravity is claimed	False reading	Those are explicit nonclaims	Definition 82.1

F Claim-Status and Reportability Matrix

Claim	Status	Carrier	Boundary
Kernel dependency	Imported/restated	Target-adequacy theorem	Generates no physical ontology
RPI role architecture	Imported	RPI theorem-use package	Role-level closure, not selected dynamics
Gravity necessity	Imported	Regular-stratum AER theorem	Fixed comparison class
Einstein normal form	Imported	Faithful metric projection with closed-source type	No source existence or PDE solution by itself
Schrödinger normal form	Imported	Projective standing, Wigner–Stone, spatial exhaustion	Closed scalar nonrelativistic scope
Attachment-safe source-bearing subtype	Accepted	Module A v1.1.0 and positive re-review	Not every response is source-bearing

Claim	Status	Carrier	Boundary
Distributional metric-source map q_g	Accepted	Structured source quotient $\widehat{Q}_{\text{src}}$	Distributional; smoothness requires extra regularity
Ward/divergence correspondence	Accepted	Attached witness and tensor distribution	Closed only when certified $J = 0$
Finite composite-source closure	Accepted	Exact term registry and balanced signed incidence	Finite completed certificates only
Typed common-response span	Accepted	$\widehat{Q}_{\text{mass}}^{\text{op}}$ with attached operator EMT, q_H , q_g , and one Casimir line	Stable scalar matching envelope only
Universal inertial-active proportionality	Accepted	Module B common charge, zero-transfer readout, contraction, and naturality	Exact stable scalar matching/calibration envelope
Upstream RPI continuation	Accepted	Module C fixed-clock Moore-history category and smooth nonidentity seed	Local regular RPI chart; no kernel-generated dynamics
Four-role projection intertwining	Accepted	Module C complete fibres, strict/declared-open processes, and cube	Exact eligibility and scope certificates required
Directed exchange and sixteen-vertex cube	Accepted	Module C common incidence, role restrictions, and action-category cocycles	Fixed-background declared-open example is not interacting quantum gravity
Lawful tensor completion and colimit	Accepted	Module D covariant cosieve completion and L_D^0	Exact source-locked Modules A–C target only
Complete joint probe and maximal skin kernel	Accepted	Module D induced probes and S_D^0	Same-target lawful equivalence only
Eight-slot certificate inhabitation	Accepted	Module D target-relative fibre finality and compatible bijection	Different physical parameter points are different target instances
Unique assignment orbit and realization licensure	Accepted	$ \Sigma_D^{gg}/\Gamma_D^{\text{lawful}} = 1$ plus Role-Occupancy Closure	Certificate-relative, not uniqueness of all richer theories

Claim	Status	Carrier	Boundary
Conditional weak equivalence	Correct conditional theorem	TB1–TB11 plus Geroch–Jang/geodesic uniqueness	No universal instantiation, SEP, or self-force
Strong equivalence, all QG, nonzero new interaction, numerical constants	Nonclaims	No carrier in v1.6	Outside frozen endpoint

G Scope Boundary and Extension Program

The fixed certificate is intentionally strong but bounded. The following extensions are rational next targets and are not deficiencies in the theorem proved here.

G.1 Strong equivalence and self-gravity

A strong-equivalence extension must treat self-gravitating bodies, gravitational binding contribution, local nongravitational experiment independence, and higher-order motion. It requires a body-family theorem stronger than the core Geroch–Jang bridge and a source ledger controlling self-field energy and gravitational radiation.

G.2 Self-force, spin, and multipoles

A Gralla–Wald-style certificate would add scaled metric families, near-zone/far-zone smoothness, first-order perturbations, center-of-mass definition, spin, multipole, and self-force ledgers. These are natural refinements of C_{trans} and C_{match} .

G.3 Locally covariant quantum field source extraction

A metric–QFT extension requires a locally covariant field or net carrier, state class, renormalized stress tensor, local covariance, anomaly control, record/update status, and divergence closure or exchange. A Yang–Mills local-net theorem on a fixed flat background is not by itself that construction.

G.4 Nontrivial interaction dynamics

The current cross-response ledger proves common source descent and projection consistency. A nonzero new interaction theorem must construct an additional standing-bearing cross-response, prove its constraint and anomaly closure, and show recovery of the current certificate in a decoupling or restriction limit.

G.5 High-curvature and horizon endpoints

A horizon extension must supply exterior witness algebra, leakage/radiation, boundary-to-boundary certificate, non-sink quotient, or a declared richer endpoint theory. Compatibility alone cannot carry information transport.

G.6 Parameter and numerical closure

The certificate does not compute absolute masses or coupling values. Those require sector-specific scale anchors, exactness certificates, quotient-stable ratios, and empirical calibration discipline. MGN1 remains the numerical-standing firewall.

H Source Map and Dependency Index

Source	Principal result used	Dependency position
Kernel [9]	Adm/Stand/Ref/Irr necessity	Root governance
Unus [10]	Singleton admissible interior	After kernel, before projections
ATS [34]	Anchor/tensor/skin trichotomy	Fixed-domain consequence layer
RPI [11]	Coherent, role-closed, minimal physical interior	Kernel-to-physics specialization
AER/common structure [13]	Quotient/constraint rigidity	Physical interior to gravity role
Gravity [12]	Gravity as structural necessity; certificate program	Before metric projection
Einstein [14]	Metric normal form and source-extraction type	Downstream metric bookkeeping
Quantum/Schrödinger [15]	Projective/Hilbert normal form and inertial representation	Downstream quantum bookkeeping
Mass/R9 [19]	Deformation response and inertial restriction	Upstream of source and kinetic representations
MGN1 [20]	Witness/quotient and no-selector discipline	Numerical standing guardrail
Records [21]	Record quotient and fixation	Boundary theorem-use package
Entanglement [22]	Compatibility-not-transport	Boundary theorem-use package
Structural unification [28]	Singleton paired projection and joint-ledger necessity	Specification before constructive modules
Module A v1.1.0 [1]	Attachment-safe source subtype, tensor-distribution representation, Ward/divergence, finite composite closure	After RPI/mass and metric licensing; before Module B
Module A positive review [2]	Independent authorization	Module A import gate
Module B accepted import [3]	Operator/source attachment, total-EMT charge, active read-out, Wigner contraction, common Casimir line, one factor, calibrated equality	After Module A; before Module C
Module B positive Cycle-4 review [4]	Mathematical authorization and typesetting-only patch authorization	Module B import gate
Module C accepted import [5]	Fixed-clock RPI continuation, complete four-role fibres, intertwiners, directed exchange, cube, strict/open inhabitants	After Modules A–B; before Module D

Source	Principal result used	Dependency position
Module C positive Cycle-3 review [6]	Independent authorization and two editorial clarifications	Module C import gate
Module D v1.4.0 [7]	Covariant cosieve completion, tensor colimit, joint probe, skin kernel, fibre exhaustion, live residue, one orbit, licensure	After accepted Modules A–C; final certificate-completion step
Module D positive Cycle-5 review [8]	Independent authorization of the Module D endpoint	Final Module D import gate
Role occupancy [27]	One compatible assignment orbit licenses realization under complete certificate hypotheses	Applied only after Module D completeness
Geroch–Jang [29]	Localized conserved SDEC tensor family implies timelike geodesic	Conditional downstream test-body branch

I Proof-Gate Closure Ledger

The frozen endpoint has not changed. Independently accepted Modules A–D close the complete certificate-relative core endpoint in their exact scopes. The test-body branch remains a separate conditional implication.

Gate	Burden	Accepted closure evidence	v1.6 status
G1	Complete certificate inhabitation	Module D constructs tensor and skin targets, proves exact finality of all eight target-relative fibres, and supplies a total pointwise eligible globally compatible bijection onto the independently formed live residue.	Closed
G2	Common-response factorization	Module A supplies attachment-safe tensor-source descent; Module B proves the intrinsic active zero-transfer and inertial scalar branches factor through one attached stable-sector Noether/Casimir charge.	Closed in the frozen stable scalar matching envelope
G3	Universal source naturality/proportionality	Module B proves naturality, one projection-wide factor α_D , no unwitnessed species-indexed conversion, flat calibration transport, and common calibrated equality.	Closed in the frozen stable scalar matching envelope
G4	Composite-source closure	Module A proves exact finite signed-incidence cancellation and Ward closure for completed source certificates, including binding, field, interaction, stress, restoration, transport, baseline, and boundary terms.	Closed within Module A scope

Gate	Burden	Accepted closure evidence	v1.6 status
G5	Upstream continuation commutation	Module C constructs the fixed-clock RPI history category first, proves lawful clock-presentation equivalence, complete four-role fibres, metric and quantum intertwiners, record and compatibility descent, directed exchange, and strict/declared-open sixteen-vertex cubes.	Closed within Module C scope
G6	Test-body hypothesis fidelity	TB1–TB11 expose every analytic, support, conservation, energy, localization, force, and correction premise. The implication to local geodesic universality remains correct conditionally.	Closed conditionally
G7	Unique complete assignment orbit	Module D proves candidate, quotient, role-set, eligibility, compatibility, failure, tensor/skin completion, live-residue identification, lawful automorphism closure, and one final assignment orbit, then applies Role-Occupancy Closure.	Closed

The core FEP-1.0 realization endpoint is reportable relative to the exact frozen target. G6 is not converted into an unconditional empirical assertion; it remains a conditional downstream theorem under TB1–TB11.

References

- [1] Amos Jay Maley. *Operational Metric-Source Representation and Composite Ledger Closure under AASC: A Distributional Stress-Energy Theorem from Complete Metric-Response Witnesses, Quotient Descent, and Finite Internal-Exchange Cancellation*. AASC-GRQM-MA-SRC-1.0, v1.1.0-attachment-corrected; manuscript SHA-256 b12ca804f79b881afebbf2ce492b11ac20293d37d2ecb4de454427cc6301cf54. 2026.
- [2] Independent hostile-referee audit. *Bounded Independent Module A Re-Review under AASC-GRQM-MA-SRC-1.0*. Cycle 2 verdict: source admission/representation correct; Ward closure correct; composite closure correct; software correspondence correct; overall Module A correct. Report SHA-256 34054b7b3ab3cde2f88d9d82ab93cff3eb76ab3c61b265369a4b6988b5578bc7. 2026.
- [3] Amos Jay Maley. *Common Mass-Charge Matching and Universal Inertial-Active Proportionality under AASC: A Poincare-Noether Form-Factor-to-Bargmann Overlap Theorem from Accepted Metric-Source Representation, Exact Zero-Transfer Ward Normalization, Complete Composite Closure, and an Explicit Wigner Contraction*. AASC-GRQM-MB-MASS-1.0, v1.3.0-normalized-ray-source-lift-corrected with authorized Cycle-4 four-command typesetting patch; corrected import PDF SHA-256 23ea7bce0926a0c4394ba65c438451dfe977aebfd2cdb53fdb1a4f6bbfb57468. 2026.
- [4] Independent hostile-referee audit. *Bounded Independent Module B Cycle-4 Re-Review under AASC-GRQM-MB-MASS-1.0*. Mathematical endpoint correct; exact four-command TikZ-CD typesetting patch authorized without further mathematical review. Report SHA-256 eac130b031670c13aaa2fac886f13d1fea606e6289fd5561f444750475de919a. 2026.
- [5] Amos Jay Maley. *Kernel-Native Upstream Continuation and Projection Intertwining under AASC: A Fixed-Clock RPI Semiflow Theorem with Lawful Metric-Orbit Finality, Clock-Conditioned Full-Ray Quantum Descent, Einstein and Schrödinger Intertwining, and Distinct Closed/Exchange-Extended Cubical Coherence*. AASC-GRQM-MC-CONT-1.0, v1.2.1-editorial-clarified-import-authorized; import PDF SHA-256 cbd8fd2c5eb9a5b937b8b02358351e3ca071fffb76a0620d6f355a57f96205a4. 2026.
- [6] Independent hostile-referee audit. *Bounded Independent Module C Cycle-3 Re-Review under AASC-GRQM-MC-CONT-1.0*. Overall standalone endpoint correct and authorized for capstone import; two non-fatal editorial clarifications implemented in v1.2.1. Report SHA-256 150041aa3e744cbd9926bd36da3786ac110170913cb6e0d627de8dbd50697b0a. 2026.
- [7] Amos Jay Maley. *Exact Certificate-Fibre Exhaustion and Unique Role-Occupancy Closure under AASC: An Eight-Sorted Completion Theorem for the Constraint-Native Gravity-Quantum Capstone*. AASC-GRQM-MD-OCC-1.0, v1.4.0-cosieve-support-corrected; manuscript SHA-256 5fb065d605900826806ac1e26d10a431701207563d16c916c2132607a421d00b. 2026.
- [8] Independent hostile-referee audit. *Strictly Bounded Independent Module D Cycle-5 Re-Review under AASC-GRQM-MD-OCC-1.0*. Overall standalone endpoint correct and authorized for capstone import. Report SHA-256 72e1c468610a42c227f2849e48d9bde13179a13067ae8d37796ae2aa65ceb1ba. 2026.

-
- [9] Amos Jay Maley. *Non-Degenerate Construction and the Kernel of Admissibility: An Exhaustion Theorem for Standing, Reference, and Irreversibility*. 2026. DOI: 10.5281/zenodo.19324199.
 - [10] Amos Jay Maley. *Unus Solus Possibilis Est: Corpus Closure and Uniqueness of the Admissible Interior*. 2026. DOI: 10.5281/zenodo.18518484.
 - [11] Amos Jay Maley. *The Realized Physical Interior: Locality, Metric Role, Fields, Particles, Gauge Structure, Mass, Vacuum, Time, Causality, and the Standard Physics Stack under AASC*. 2026. DOI: 10.5281/zenodo.20129729.
 - [12] Amos Jay Maley. *Gravity as Structural Necessity: An AASC Account of the GR-Type Physical Interior*. 2026. DOI: 10.5281/zenodo.19945401.
 - [13] Amos Jay Maley. *Admissibility and Constraint-Induced Rigidity of Physical Quotient Structure: GR and Quantum Mechanics as Bookkeeping Projections of an AASC Constraint Architecture*. 2026. DOI: 10.5281/zenodo.18630901.
 - [14] Amos Jay Maley. *Einstein Dynamics as the Unique Minimal First-Response Metric Projection of AASC Gravity*. 2026. DOI: 10.5281/zenodo.20538579.
 - [15] Amos Jay Maley. *Schrödinger Dynamics as the Unique Minimal First-Response Hamiltonian Projection of AASC Quantum Standing*. 2026. DOI: 10.5281/zenodo.21254589.
 - [16] M. H. Stone. “On One-Parameter Unitary Groups in Hilbert Space”. In: *Annals of Mathematics* 33.3 (1932), pp. 643–648. DOI: 10.2307/1968538.
 - [17] Eugene P. Wigner. *Group Theory and Its Application to the Quantum Mechanics of Atomic Spectra*. Academic Press, 1959.
 - [18] Valentine Bargmann. “On Unitary Ray Representations of Continuous Groups”. In: *Annals of Mathematics* 59.1 (1954), pp. 1–46. DOI: 10.2307/1969831.
 - [19] Amos Jay Maley. *Mass, Inertia, and Persistence Compression: The Deformation-Resistance Role of Particle Identity under Field-Carrier Mediation*. AASC corpus manuscript. 2026.
 - [20] Amos Jay Maley. *Deformation-Response and Numerical Standing over the SR7 Skeleton*. MGN1, Foundations for the Mass-Generation and Numerical-Readiness Arc. 2026.
 - [21] Amos Jay Maley. *Admissible Record Construction in Physically Non-Selective Quantum Regimes: Fixed-Domain Quotients, Boundary-Trace Fixation, and Constructional Determinacy*. 2026. DOI: 10.5281/zenodo.18514647.
 - [22] Amos Jay Maley. *Quantum Entanglement as Boundary-Level Compatibility: Unique Same-Domain Admissible Classification on the Bell Scope*. 2026. DOI: 10.5281/zenodo.19401432.
 - [23] J. S. Bell. “On the Einstein Podolsky Rosen Paradox”. In: *Physics Physique Fizika* 1.3 (1964), pp. 195–200. DOI: 10.1103/PhysicsPhysiqueFizika.1.195.
 - [24] Amos Jay Maley. *The Structure of Coherent Construction: A Bivalent Trajectory Theory*. AASC corpus manuscript. 2026.
 - [25] Amos Jay Maley. *Connection as Standing-Preserving Transport Geometry*. Gauge Realization Rigidity Arc manuscript. 2026.
 - [26] Amos Jay Maley. *Covariant Differentiation from the AASC Kernel*. AASC corpus manuscript and audit supplement. 2026.

-
- [27] Amos Jay Maley. *Role-Occupancy Closure under Admissibility*. AASC corpus manuscript. 2026.
 - [28] Amos Jay Maley. *Structural Unification of Gravity and Quantum Dynamics under AASC: Einstein and Schrödinger Dynamics as Forced Paired Projections of the Admissible Physical Interior*. AASC corpus manuscript. 2026.
 - [29] Robert Geroch and Pong Soo Jang. “Motion of a Body in General Relativity”. In: *Journal of Mathematical Physics* 16.1 (1975), pp. 65–67. DOI: 10.1063/1.522416.
 - [30] James Owen Weatherall. “A Brief Remark on Energy Conditions and the Geroch–Jang Theorem”. In: *Foundations of Physics* 42.2 (2012), pp. 209–214. DOI: 10.1007/s10701-011-9606-0.
 - [31] Jürgen Ehlers and Robert Geroch. “Equation of Motion of Small Bodies in Relativity”. In: *Annals of Physics* 309.1 (2004), pp. 232–236. DOI: 10.1016/j.aop.2003.08.020.
 - [32] Samuel E. Gralla and Robert M. Wald. “A Rigorous Derivation of Gravitational Self-Force”. In: *Classical and Quantum Gravity* 25.20 (2008), p. 205009. DOI: 10.1088/0264-9381/25/20/205009.
 - [33] Robert Geroch and James Owen Weatherall. “The Motion of Small Bodies in Space-Time”. In: *Communications in Mathematical Physics* 364.2 (2018), pp. 607–634. DOI: 10.1007/s00220-018-3268-8.
 - [34] Amos Jay Maley. *Anchor, Tensor, and Skin: A Fixed-Domain Decomposition Theorem for Admissible Description*. 2026. DOI: 10.5281/zenodo.18678041.